

**A WEB-BASED ORDERING AND RESERVATION SYSTEM
FOR 4KS ENTERPRISE AT KABACAN,
NORTH, COTABATO**

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INTRODUCTION

Technology has revolutionized the way businesses operate, it offers efficient tools for management, marketing, and customer engagement. Small businesses the backbone of many communities, that provides essential goods, services, and employment opportunities. Continue to rely on personalized customer relationships, local market knowledge, and adaptability to thrive. Small businesses contribute significantly to economic development, foster innovation and support local economies. They also preserve unique products and services that may not be available through larger chains. For many entrepreneurs, a small business is more than just a source of income. It represents passion, hard work, and community impact. As such, effective management strategies are vital to overcoming challenges and ensuring long-term success in a competitive market.

4ks Enterprise is a dynamic small business dedicated to producing customized products with a strong focus on customer satisfaction. It specializes personalized items including tarpaulins, custom mugs, printed t-shirts, lanyards, trophies, medals, and more. The business caters to both organizations and individual clients across Matalam and Kabacan, North Cotabato. Founded in 2006, 4ks Enterprise began its operations in Poblacion, Matalam, and expanded in 2015 with a second branch in Poblacion, Kabacan . With a dedicated team of 11 employees, the enterprise prides itself on creativity, efficiency, and

commitment to exceptional service, ensuring that each product meets the unique needs of its valued customers.

A well-structured business management system is crucial for organizing operations, tracking finances, and ensuring sustainable growth. Currently, the problems facing by the 4ks Enterprise is the order are managed through direct communication between customers and employees via mobile phone or social media, with some customers visiting the shop in person to place or update orders. Order details, including product specifications and delivery schedules, are recorded manually that leads to inconsistencies in tracking due to the absence of centralized system. Since there is no centralized system for scheduling and monitoring orders, staff occasionally forget customer requests, resulting in delayed deliveries. Furthermore, the lack of a standardized scheduling process sometimes causes order mix-ups, such as producing items for incorrect dates.

To address these problems, this study proposed the development of a Web-Based Ordering and Scheduling System for 4ks Enterprise. The system aims to streamline the order management process by allowing customers to place and monitor orders online. It include features for order tracking, scheduling, customer notifications, and database management to ensure accurate and efficient handling of transactions.

Significance of the Project

The goal of this project is to develop a Web-Based Ordering System for 4ks Enterprise that will streamline the order management process, reduce errors, and improve efficiency for both the business and its customers. The system will feature automated order tracking, real-time updates, inventory management, and sales reporting.

This will benefit:

Customers will enjoy a convenient ordering experience, allowing them to place orders anytime without the need to call or visit the store. They'll also receive timely updates on their order status and pickup notifications.

4ks Enterprise, as the system will benefit from the elimination of lost or mixed-up orders, improved inventory tracking, and provide valuable sales insights to support better decision-making.

Employees will no longer need to record orders manually, reducing the chance of errors and minimizing their workload. The centralized system will keep everything organized, making their tasks more manageable and efficient.

This system will save time, reduce mistakes, and enable 4ks Enterprise to operate more effectively. It is a practical innovation that can inspire other small businesses to embrace digital solutions for growth and sustainability.

Statement of the Problem

At present, 4ks Enterprise manages customer orders manually, relying on phone calls, social media messages, and in-person visits due to the absence of a digital ordering system. This outdated approach often leads to miscommunication, delays, and errors that negatively affect both the business and its customers. Orders are manually recorded, increasing the risk of forgotten requests or misplaced information. Additionally, the lack of an automated scheduling system results in production mix-ups and incorrect order fulfillment.

Longstanding clients frequently encounter issues when placing orders. Messages sent via SMS or social media often go unanswered for extended periods, and orders are frequently not ready by the promised pickup dates. The business also faces challenges with inefficient tracking methods, as there is no centralized system to monitor order statuses in real time, prioritize production, or maintain accurate customer records for repeat order transactions.

Objectives of the Project

The general objective of this study is to develop a Web-Based Ordering and Reservation System for 4ks Enterprise.

Specifically, this study aims to:

1. develop a user-friendly web-based platform that allows customers to place orders and make reservations.
2. implement an inventory tracking system to prevent stock shortages and overstocking;
3. enable automated notifications via email for order updates and reservation confirmations;
4. develop a reporting and analytics module that generates real-time sales reports. Provide insights into customer purchasing behavior, analyzes inventory, and track reservation trends; and
5. evaluate the functionality of the system.

Expected Outputs of the Project

The expected output of this study is a fully functional web-based ordering and reservation system specifically designed for 4ks Enterprise, a small business in Kabacan, North Cotabato. For customers, it would be a simple website where they can browse the menu, place orders, and make reservations from anywhere, and they would get automatic email confirmations and updates. For the business, it includes a dashboard for staff to manage stock levels so they never run out of products, and to create instant sales reports that help analyze inventory, track customer reservations of the customer and understand what customers like and how they behave. The main goal is to test this new system thoroughly to make sure it works, so it can simplify all the weekly tasks,

cut down on manual errors, and create a much better experience for both the customers and the staff.

Operational Definition of Terms

Terms used in this study are operationally defined.

Admin - An authorized employee of 4ks Enterprise who manages the system, including inventory, order processing, and customer records, through an administrative dashboard.

Customer - An individual or organization that uses the web-based system to place orders or make reservations for products offered by 4ks Enterprise.

Customer Records - A database within the system that stores customer details to facilitate personalized service and streamline future orders.

Inventory Tracking System - A module within the system that monitors stock levels of products, generates alerts for low stock, and helps prevent shortages or overstocking.

Notifications - System-generated messages via email sent to customers and staff to confirm orders, provide updates, remind of due dates, or alert about low inventory levels.

Order Tracking - A feature of the system that allows customers and staff to monitor the status of an order, such as processing, completed, or ready for pickup or delivery.

Reporting and Analytics Module - A system component that generates reports on sales, customer purchase behavior, inventory turnover, and reservation trends to aid business decision-making.

Scheduling System - A feature that organizes order deadlines and production

timelines to prevent mix-ups and ensure timely completion of products.

User-Friendly Web Interface - A design of the system that ensures ease of navigation and usability for customers, enabling them to place orders or make reservations with minimal effort.

Web-Based Ordering and Reservation System - A digital platform accessible via the internet that allows customers to place orders and make reservations for pick up or delivery offered by 4ks Enterprise. It includes features such as order tracking, inventory management, and automated notifications.

4ks Enterprise - A small business established in 2006, specializing in customized products such as tarpaulin prints, mugs, t-shirts, lanyards, trophies, and medals, with branches in Matalam and Kabacan, North Cotabato.

Conceptual Framework

Figure 1 shows this conceptual framework, which details the system by defining its components across four stages: input, process, outcome, and output. The input includes various data types such as User Information, Order Details, Reservation Details, Payment Information, Product data, and Inventory data. The central process involves accepting custom orders, validating them, calculating costs, managing real-time reservations, monitoring inventory levels with alerts for staff, sending automated order status emails to customers, and

compiling data for management reports. The expected outcomes are key benefits: staff spending less time on manual tasks, successful customer self-service, reduction in manual errors, production of accurate reports, and increased overall efficiency and customer satisfaction. The system's output includes a Web-based platform, automated Notifications, Sales and Revenue reports, a Comprehensive database, and detailed Inventory reports for optimizing purchasing.

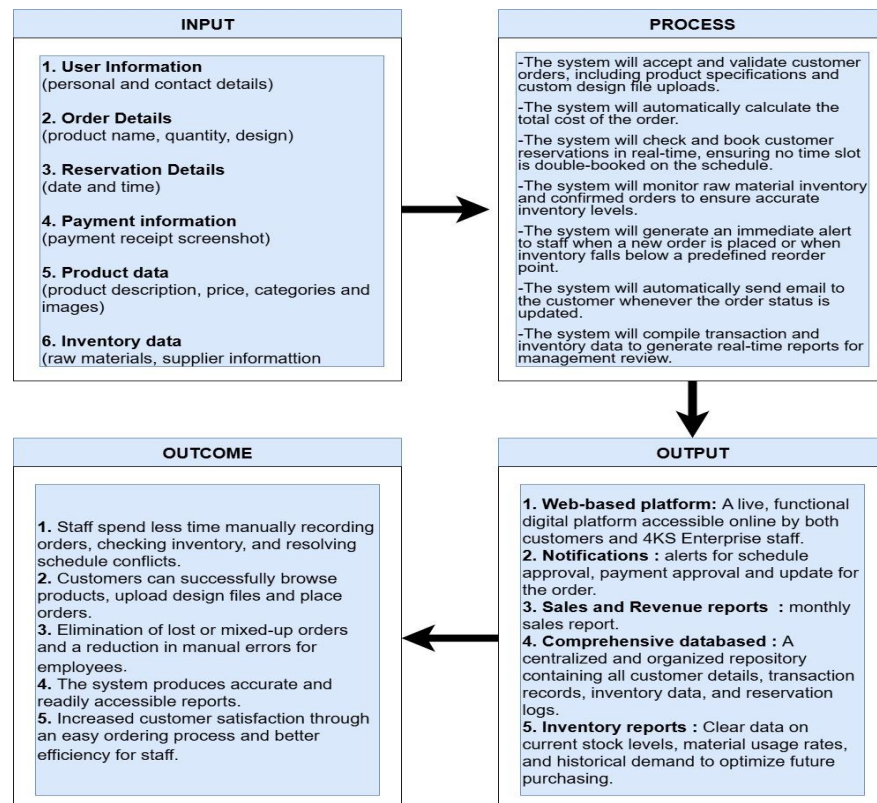


Figure 1. Conceptual Framework for Web-based Ordering and Reservation System for 4ks Enterprise

Limitations of the Project

The proposed project will focus on order management, inventory tracking, reservation scheduling, customer management, and automated notifications, all integrated into a web-based system with real-time reporting capabilities. The system administrator will manage all data and have full database access. This project is specifically designed for 4ks Enterprise, catering to its customers and employees across its branches in Kabacan and Matalam, North Cotabato.

People

The system involves customers who will place orders and reservations of dates for pickup/delivery, and 4ks Enterprise staff who will process orders, monitor inventory, and generate reports. Customers can track orders in the platform, while staff manage workflows and resolve issues. The admin role is the highest level of user access and control. The system aims to streamline operations, reducing manual errors for all users.

Data

The system will track and store order details, reservations, inventory levels, customer information, and transaction histories, ensuring accurate records and analytics. Products such as customized mugs, shirts, tarpaulins, lanyards, sashes, trophies, medals, and certificates, and Inventory for the materials to use for the products. Only registered customers and authorized staff can access the system.

Process

The staff serves as the system controller, with authority to update inventory, modify orders, monitor transactions, and generate sales reports. The web platform allows customers to browse products, place orders, and receive notifications. While the admin will manage the users of the system, independently reviewing logs of all staff activity and having operational and administrative control over the staff. For the payment process, they need to pay 40-50% the price of the product they ordered, and they need to pay through GCash. After paying, the customer needs to send the receipt of that payment to the staff account in the ordering and reservation system. The system does not include refunds, returns, or cancellations of orders.

REVIEW OF RELATED LITERATURE

As e-commerce continues to grow, small businesses are increasingly leveraging various online platforms to reach potential consumers. Given constraints such as limited technical capabilities, selecting an efficient business model becomes essential. Research indicates that small business efficiency improves with increased functional integration, as it fulfills utilitarian motivations. Platforms that combine functional integration with social presence to address motivations tend to enhance the efficiency of all participating small businesses. The development of information and communication technologies has diversified the channels available to offline sellers, enabling them to create cost-effective online stores for product promotion and sales. Strategically, establishing a presence on e-commerce platforms allows small business owners to meet consumer needs driven by both utilitarian motivations, such as convenience, information availability, trust, and time efficiency, and motivations, including social engagement and trend exploration. Consequently, utilizing diverse e-commerce platforms is a beneficial strategy for small business growth (Hana et al., 2018)

Order Management System

An order management system plays a vital role in supply chain management as it significantly influences customer satisfaction and overall company profitability. Traditional methods that rely on manual record-keeping are often prone to errors, leading to issues such as overstocking, understocking, and ultimately, missed sales opportunities. With the integration of advanced technologies such as GPS tracking systems, modern order management processes have become more efficient and accessible, allowing consumers to conveniently order ingredients online. The system aims to provide not only the list of recipe ingredients but also the exact quantities required, ensuring timely delivery within one to two hours. This technological advancement addresses both time efficiency and quantity accuracy, ultimately offering enhanced service to consumers while minimizing waste (Ilyas et al., 2021).

According to Dr. R. Sivanesan (2017), the rapid evolution of technology has significantly transformed consumer buying behavior, with many preferring the convenience of online shopping over traditional in-store purchasing. Consumers are drawn to the ease of shopping online, especially since it allows them to browse, compare prices, and make informed decisions without physically visiting multiple stores. Online shopping reduces the effort and time involved in traveling and offers access to a wider range of products and pricing options. Furthermore, the interaction between buyers and sellers in e-commerce is mostly impersonal, relying on digital communication such as chats

and emails. This shift in the nature of buyer-seller communication highlights a key difference from traditional retail settings. The customers of various age groups, particularly the youth segment of society, have exhibited a profound presence on different online sources for shopping purposes. Thus, customers' online presence for shopping purposes has increased their shopping options enormously and widened their choice (Ellison et al., 2020).

Gnanadhas (2014) emphasizes the shift in lifestyle among people in modern times, noting a growing discomfort with visiting crowded markets due to time constraints. In response to this, E-shopping has emerged as a convenient alternative, offering consumers the ability to purchase goods and services directly from sellers via the Internet without intermediary involvement. This mode of shopping allows consumers to browse online stores from the comfort of their homes or workplaces, leveraging the widespread availability of internet access.

According to Gren et al. (2019), after the system is designed, tested, and found to work as expected, it is time to go live. There are several different strategies for deploying an Order Management System. The company can switch to using the new system at once, take the new system to use gradually, or combine some of the components from each method. The deployment strategy depends mainly on the scope of the system, size of the company, the amount of data migration and the employee's ability to learn and adjust to the new system.

According to Kocaoglu and Acar (2016), Order processing is strictly related to information flows in the system and includes a number of operations. A sales order process normally consists of internal document generation within a firm with the aim of managing sales transactions. Customers may have to request the products by filling out an order form. These orders are transmitted and checked. The availability of the requested items and the customer's credit status are then verified. Later on, items are retrieved from the stock (or produced), packed, and delivered along with their shipping documentation. Finally, customers have to be kept informed about the status of their orders.

Organisational Change Management (OCM) is an inevitable and inseparable component of Enterprise Resource Planning (ERP) implementation, as it directly influences both business processes and employee behavior. ERP systems are designed to enhance overall business performance, but their introduction often leads to significant technological and procedural changes that affect employees' daily tasks. Such transitions can initially trigger uncertainty, frustration, and skepticism among end users, highlighting the need for clear communication and active leadership involvement throughout the implementation process. Effective communication, particularly with lower-level employees, helps address their concerns and builds trust and confidence in the new system. Studies reveal that end-user acceptance and adaptability are closely related; when users are able to adapt effectively, their acceptance of the system increases. Training also plays a crucial role in this process, as improving

employees' IT skills enhances their comfort and ability to use the ERP system efficiently. However, technical competence alone does not guarantee successful implementation without consistent management support and engagement (Mk & Sarukesi, 2025).

According to Snehal et al. (2024), an online food ordering system is designed to allow customers to conveniently browse restaurant menus, select items, and place orders either for delivery or self-pickup. The system enables efficient management of customer orders by restaurant staff, reducing errors and saving time. It provides real-time updates on food preparation and delivery status, ensuring transparency and improved customer experience. Technically, the system is developed using HTML and PHP for the front end and MySQL for the back end, ensuring secure and reliable data handling. It also supports multiple payment options, including cash and credit or debit cards. Overall, the ordering system enhances restaurant operations by streamlining order processing, improving accuracy, and increasing customer satisfaction through automation and accessibility.

Reservation Management System

Online reservation management systems have revolutionized how users interact with booking platforms by prioritizing convenience, efficiency, and personalization. Studies emphasize that a user-friendly interface, clear information layout, and responsive design across devices are essential in

enhancing the overall user experience. Furthermore, the integration of advanced technologies such as artificial intelligence and machine learning contributes to smarter reservation processes through features like personalized recommendations and dynamic availability updates. These innovations not only streamline operations but also ensure that users receive timely and relevant options, thereby increasing satisfaction and trust. As a result, modern reservation systems serve as vital tools in various industries, enabling seamless service access while maintaining reliability and user engagement (Deepa et al., 2023).

According to Williams and Ajinaja (2019), the advancements in mobile technology and widespread broadband access have significantly enhanced the functionality and accessibility of online reservation management systems. These systems now play a crucial role in streamlining the booking process, supporting real-time order allocation, and improving operational efficiency. Effective reservation systems integrate modern technologies such as booking engines and centralized control panels to manage reservations across multiple sales channels. This integration allows for better decision-making, dynamic reservations, and the ability to analyze each channel's contribution to overall revenue. Furthermore, the use of computerized systems supports timely information retrieval, updates, and report generation, which are essential for maintaining accurate customer records and adapting to changes such as cancellations or modifications. Transitioning from manual methods to

automated reservation systems not only reduces inefficiencies but also helps achieve commercial goals by optimizing booking procedures and enhancing customer satisfaction.

A web service is an application that resides and runs on the web and provides services to other applications over Hypertext Transfer Protocol (HTTP). In essence, a web service is a distributed application, which in basic terms comprises a service and a client. In SOA terms, a client is also referred to as a consumer or requester. Web services can be implemented within the same organization or between two business organizations trading with each other. The web service will provide an inter-enterprise service. What sets web services apart from other distributed systems is their ability to run or operate independently of the underlying IT infrastructure. In other words, a web service provides its services to other applications without paying any attention to the hardware and operating system on which those applications run or the programming language in which they are written. The use of open standards and protocols is responsible for this much-needed feature of web services (Odemba, 2019).

According to Rahmatya et al. (2020) that many organizations lack complete and accessible information online, making it difficult for customers to obtain details and make reservations. It aimed to provide adequate information, enable online reservations and payments, and serve as promotional media for the tourist site. Using the object-oriented approach and the Waterfall model, the

study covered the requirements analysis and system design phases, utilizing UML diagrams to illustrate system processes. It allows customers to view event details, check available dates, make reservations, and confirm payments through a website. Meanwhile, operational staff can manage promotions, monitor bookings, and validate payments.

Inventory Management

According to Tanaman et al. (2023), a web-based inventory management system was developed to address the inefficiencies of a manual, paper-based inventory process used by a small business. The manual method relied on paper forms for inventory tracking, which often resulted in data inaccuracies, misplaced or damaged records, and inefficient reporting. Employees frequently recorded information on the same forms, leading to duplicate entries and incorrect data. Additionally, generating accurate sales reports proved challenging due to the reliance on pen, paper, and spreadsheets. The web-based system was introduced to digitize these processes, allowing inventory data to be recorded and accessed electronically.

Otobo and Alegbe (2024) designed a web-based inventory management system specifically for small and medium-sized production companies, addressing the challenges these businesses face in tracking stock across multiple locations. Their system aimed to optimize inventory control through real-time monitoring, automatic stock level updates, and intelligent reorder point

detection, thereby enhancing operational efficiency. By implementing technologies such as Java RMI with SSL, MySQL, PHP, and HTML, the researchers created a secure and interactive platform capable of coordinating inventory data from multiple stores in a centralized database. Many businesses, particularly small and medium enterprises (SMEs), still rely on manual inventory tracking. To address these challenges, they proposed and developed a Web-based Inventory Control System designed to enhance accuracy and streamline operations. It incorporates modern tracking technologies such as QR codes and barcodes to monitor the movement of goods throughout the supply chain from the point of purchase to sales, between store locations, and within individual outlets. It provides up-to-date inventory information to the corporate office and supports remote access for key personnel, allowing for improved coordination and decision-making. Their study highlights the system's effectiveness in reducing the time spent on inventory tasks, improving stock visibility, and enabling better oversight of multi-store operations, ultimately leading to improved customer satisfaction and increased profitability (Erameh & Odoh, 2021).

User interface design is the specification of a dialogue between the system user and the computer. The user interfaces were developed as HTML and PHP-based interfaces, with a menu-driven strategy that requires the user to choose an action from a menu or a list of alternatives (Adams, 2023).

According to Soegoto and Palalungan (2020), Inventory refers to the goods or resources managed within a company. Many companies utilize web-based online inventory systems because these provide a centralized database accessible through the internet, allowing inventory managers to efficiently communicate with employees and access accurate, real-time information. Web-based inventory systems offer several advantages, including saving time, reducing effort and costs, easing stock monitoring, and facilitating report generation and stock replenishment decisions. Developed in alignment with business process flows, these systems enhance operational efficiency by integrating smoothly into company workflows. The effectiveness of inventory management is typically measured by a company's ability to provide high-level customer service, maintain low inventory investment, maximize throughput, and minimize costs. Additionally, inventory systems aim to meet anticipated customer demand and maintain customer satisfaction by separating various production components, which helps prevent fluctuations in the manufacturing process by buffering production operations with additional inventory. Manual inventory management using spreadsheets is often inefficient leading to inaccurate stock records, excessive paperwork, and difficulties in tracking sales and updating inventory in real time. To address these challenges, a computerized, web-based inventory management system is essential. Such a system automates key processes including stock tracking, sales recording, employee salary management, and order handling. Web-based systems offer

the advantage of storing data in centralized databases that can be accessed quickly through the internet or intranet, making inventory information readily available for better decision-making (Agboola et al., 2022).

Aswanth KK et al. (2019) developed a web-based inventory and ERP system for ACME Pumps designed to automate and streamline the company's operations. The system integrates major business functions such as product management, sales, billing, and reporting into a single, centralized platform. One of its key features is the automatic stock alert system, which notifies the administrator through SMS and email when inventory levels drop below the minimum threshold. It also generates various reports, including sales, customer, invoice, and stock reports, allowing for quick and accurate data retrieval. The proposed system was developed using Microsoft C# .NET as the front-end and SQL Server as the back-end, ensuring efficient data handling and user-friendly operation. It eliminates the drawbacks of the previous manual system, which was time-consuming, prone to errors, and difficult to update. By consolidating operations into one database and interface, the ERP-based system enhances data accuracy, reduces paperwork, and improves business process efficiency. The inclusion of modules such as admin authentication, customer management, item tracking, production and sales details, and reporting provides better control, security, and real-time monitoring. Overall, the study highlights that the adoption of an automated ERP inventory system significantly improves

accuracy, productivity, and decision-making, while minimizing manual effort and human error.

METHODOLOGY

This chapter outlines the systematic approach used in developing and evaluating the Web-Based Ordering and Reservation System for 4ks Enterprise. It presents the research design, criteria for selecting participants, and data collection methods to ensure accuracy and reliability, while also considering ethical practices. Furthermore, it details the system development framework, including the project timeline using a Gantt chart, technical specifications, and core functional requirements essential for successful implementation.

Research Design

The researcher will apply a descriptive research design to assess the functionality of the proposed system. Data will be gathered through surveys and interviews with key stakeholders, specifically the staff and selected customers of 4Ks Enterprise.

Research Participants

The study will include twenty-seven (27) participants, consisting of seven (7) staff and twenty (20) customers of 4ks Enterprise for both branches, Matalam and Kabacan branches. The staff will evaluate the system's

administrative functions, while the customers will test the user interface along with ordering, scheduling, and notification features. Their feedback will be used to assess the system's functionality.

Research Instrument

A system will be developed, which will be tested and evaluated by identified users. The researcher will use a set of survey questionnaires to collect data for assessing the system's functionality.

Data Collection

The researcher will submit a formal request to the 4ks Enterprise office to conduct interviews. After approval, the researcher will interview the business owner to gather essential data and information for the system's development. In addition, a survey questionnaire will be given to the selected customers to assess the system's functionality.

Data Analysis Procedure

The collected data will be analyzed using descriptive methods to assess the system's functionality and efficiency. Responses will be grouped and categorized, then summarized through descriptive statistics such as frequency,

percentage, and mean scores. The results will be interpreted using the values presented in Table 1 to ensure alignment with the project's objectives.

Table 1 . Likert Scale for Evaluating Systems Functionality.

Numerical Value	Description	Weighted Average
1	Strongly Disagree	1.00-1.49
2	Disagree	1.50-2.49
3	Agree	2.50-3.49
4	Strongly Agree	3.50-4.00

Validity

To make sure the study is valid, all processes, features, and updates in the Web-Based Ordering and Reservation System will be clearly written down. The researcher will follow proper research steps and talk directly with staff and selected customers to check if the information gathered is correct. All data will be carefully recorded and reviewed to avoid mistakes and keep it accurate. The researcher will stay fair by avoiding personal opinions and will only present facts that are confirmed by participants. The results will be shown in a clear and simple way to explain the system's impact. Keeping detailed records of how the data was collected will also help prove the study is trustworthy and reliable.

Ethical Considerations

To keep participants' information private, no personal details will be included. The researcher will make sure that all collected data stays confidential

and can only be accessed by the researcher and authorized people in the project. For security, the system will have a login with a username and password so that only authorized users can view sensitive information. Before collecting data, the researcher will ask permission from staff and customers, clearly explaining the purpose, goals, and any possible risks or benefits of the study. Participants will also be told that joining is voluntary and that they can stop at any time without pressure or negative effects.

Project Development

Design of the Project

Figure 2 shows the Waterfall model. To ensure a systematic and organized development process for the Web-Based Ordering and Reservation System, the researcher will adopt the Waterfall model. This model follows a step-by-step process where each phase is finished before starting the next. It provides a clear structure, reduces risks, and ensures that all system requirements are fully met.

The process starts with the Requirements Analysis phase, where the researcher carefully examines how customers and staff currently manage orders and schedules. During this stage, issues such as forgotten orders, mixed-up pick-up dates, and incorrect orders are identified. Recognizing these

problems is crucial as it helps establish clear objectives and determine the essential features and functionalities that the new system must provide.

Once the requirements are clear, the process moves to the System Design phase, which focuses on creating the overall system architecture, database structure, wireframes, and user interface layouts for both the admin, staff, and customer dashboards. At this point, the workflows for key processes such as order processing, reservation handling, and report generation are carefully defined to ensure clarity and efficiency. In addition, security measures, including authentication protocols and data protection, are incorporated to maintain data integrity and safe user information.

Once the design is finalized, the project proceeds to the Development phase, where the system is constructed based on the approved specifications. During this stage, the developer writes the necessary code, builds the web-based platform, and integrates essential modules such as Order Management, Reservation Management, and generates reporting. The interface is also carefully developed to be user-friendly and intuitive, ensuring that both staff and customers can navigate and utilize the system with ease.

In the Testing phase, the system is evaluated through functionality to detect and fix issues. Feedback from test users helps refine the system, ensuring all modules work properly and data remains secure.

Finally, the project moves to the Implementation phase, where the system is deployed in a live environment and made fully operational. This stage

involves setting up hosting, configuring the system, and providing training to staff to ensure they can confidently and effectively utilize the system's features.

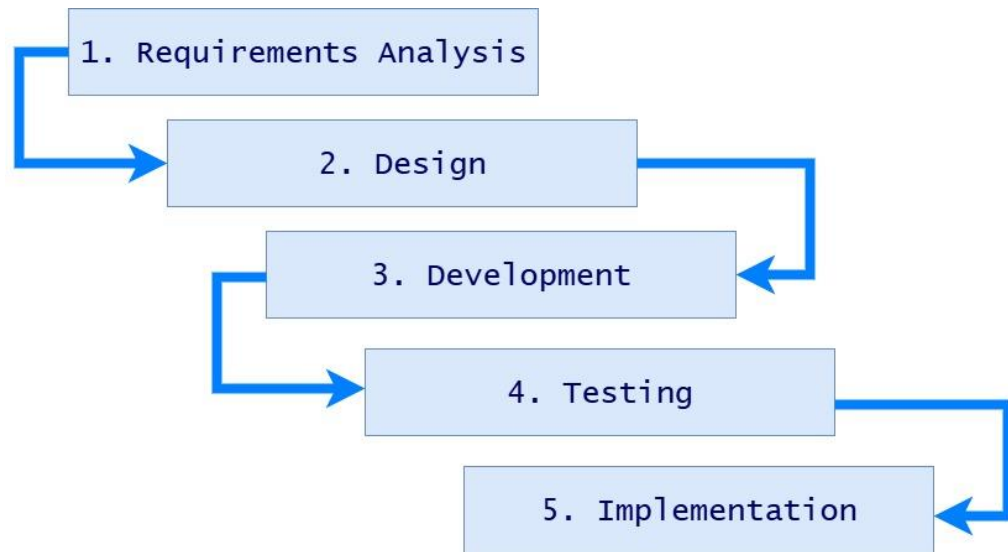


Figure 2. Waterfall Model

System Requirement Specification

Functional Requirements

- **Account Registration**

Input – User details.

Process – Validate and store user data in the database.

Output – User account created and confirmation message.

- **Edit User Profile**

Input – Updated user information

Process – Verify and update user details in the database.

Output – Success message and updated profile displayed.

- **Inventory Management**

Input – Product details.

Process – Track stock levels, trigger low-stock alerts

Output – Inventory report, restock alerts

- **Login**

Input – Username/email and password.

Process – Authenticate credentials

Output – Access granted, redirect to dashboard or error message.

- **Make Reservation**

Input – Preferred date or time, purpose.

Process – Check availability, store reservation in the database

Output: Reservation confirmation and reminder notification

- **Manage User Accounts**

Input – Admin actions.

Process – Apply changes to user accounts in the database.

Output – Updated user list or confirmation message.

- **Place Order**

Input -Product selection, customization details, delivery/pickup preferences

Process – Validate order, check inventory, store order in the database

Output – Order confirmation, automated notification email, order ID

- **Send Notifications**

Input – Trigger event.

Process – Generate and dispatch email alerts

Output – Notification sent to customer or staff

- **System Evaluation**

Input – Performance metrics.

Process – Analyze data, generate insights

Output – Evaluation report with KPIs and recommendations

- **Track Order Status**

Input – Order ID or user session

Process – Retrieve order details from the database

Output – Order status and history

- **Update Order (Staff)**

Input – Order ID, new status.

Process - Validate changes, update the database, trigger notifications

Output – Success message and updated order status

Software Requirements Specification

Table 2 details the software requirements specification for the system, specifying both the minimum and recommended requirements needed to ensure compatibility with the proposed project.

Table 2. Software Requirement Specification

Specification	Minimum	Recommended
Operating System	Windows 10	Windows 11
Database Server	MySQL	MySQL
Programming Application	VS Code	VS Code
Programming Language	Php	Python, Javascript

Hardware Requirements Specification

Table 3 outlines the hardware requirements specification for the system, detailing both the minimum and recommended hardware specifications necessary to ensure the proposed project can be properly integrated and is fully compatible.

Table 3. Hardware Requirement Specification

Specification	Minimum	Recommended
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4GB	16GB or higher
Storage	120GB HDD	1TB HDD/SSD or higher
Backup Drive	External USB	Cloud Backup

Project Design

User Interface Design

Figure 3 shows a login interface for the 4ks Enterprise Ordering and Reservation System, featuring roles which are customer, staff, and admin, username and password fields, a login button, and an option to register for new users.

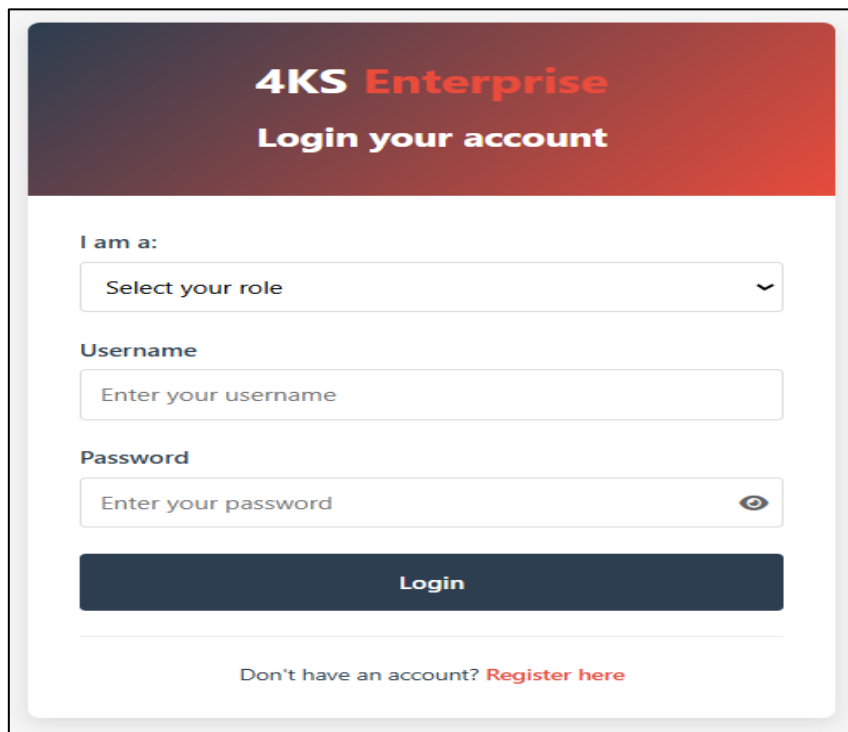
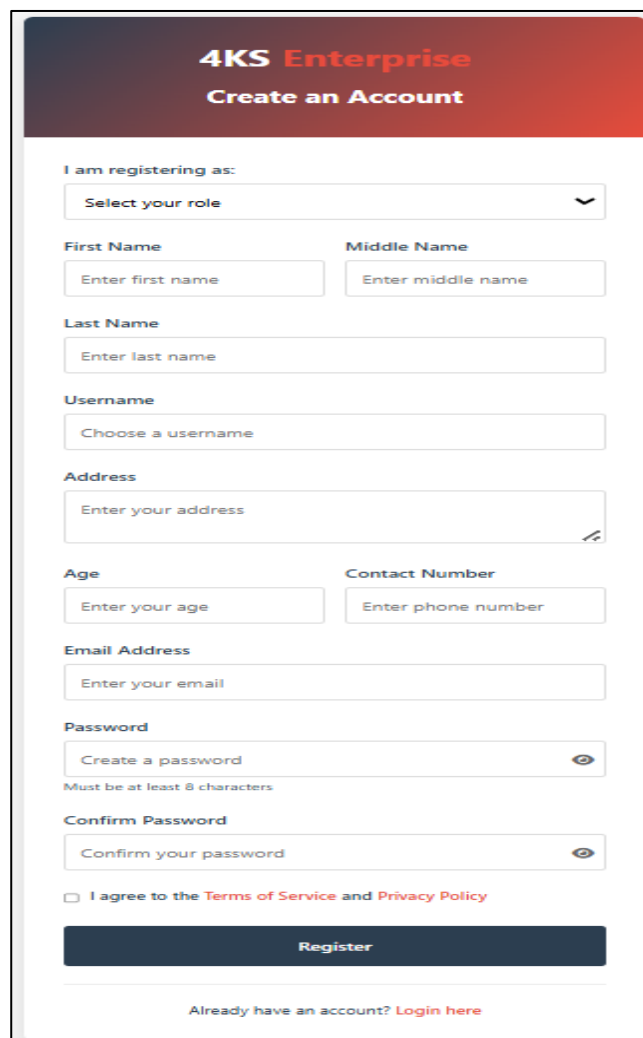
The image shows a login form for '4KS Enterprise'. At the top, there is a dark red header with the text '4KS Enterprise' in white and 'Login your account' in a smaller white font. Below the header, the form is white with a light gray border. It contains a dropdown menu labeled 'I am a:' with the text 'Select your role' and a downward arrow. Below this is a text input field labeled 'Username' with the placeholder text 'Enter your username'. Underneath is another text input field labeled 'Password' with the placeholder text 'Enter your password' and a small eye icon to its right. A dark blue button with the text 'Login' is positioned below the password field. At the bottom of the form, there is a link that says 'Don't have an account? Register here' in a small, dark blue font.

Figure 3. Login for Web-based Ordering and Reservation System

Figure 4 shows the registration page for the 4ks Enterprise system. This form requires users to select their role and input personal information, including full name, username, address, age, contact number, email address, and password. The form also includes a checkbox for agreeing to the Terms of Service and Privacy Policy before the user can successfully register a new account.



The registration form for the 4KS Enterprise system is displayed within a red header bar. The header contains the text "4KS Enterprise" in white and "Create an Account" in white. Below the header, the form is organized into several sections. The first section is labeled "I am registering as:" and contains a dropdown menu with the text "Select your role". The second section is labeled "First Name" and "Middle Name" and contains two input fields with placeholder text "Enter first name" and "Enter middle name". The third section is labeled "Last Name" and contains an input field with placeholder text "Enter last name". The fourth section is labeled "Username" and contains an input field with placeholder text "Choose a username". The fifth section is labeled "Address" and contains an input field with placeholder text "Enter your address". The sixth section is labeled "Age" and "Contact Number" and contains two input fields with placeholder text "Enter your age" and "Enter phone number". The seventh section is labeled "Email Address" and contains an input field with placeholder text "Enter your email". The eighth section is labeled "Password" and contains an input field with placeholder text "Create a password" and a small eye icon. Below the password field is the text "Must be at least 8 characters". The ninth section is labeled "Confirm Password" and contains an input field with placeholder text "Confirm your password" and a small eye icon. Below the confirm password field is a checkbox with the text "I agree to the Terms of Service and Privacy Policy". At the bottom of the form is a dark blue button with the text "Register". Below the button is a link that says "Already have an account? Login here".

Figure 4. Registration for Web-based Ordering and Reservation System

Admin User Interface Design

Figure 5 shows the Admin Dashboard of the 4ks Enterprise system. This interface allows the system administrator to manage users, products, and view system reports.

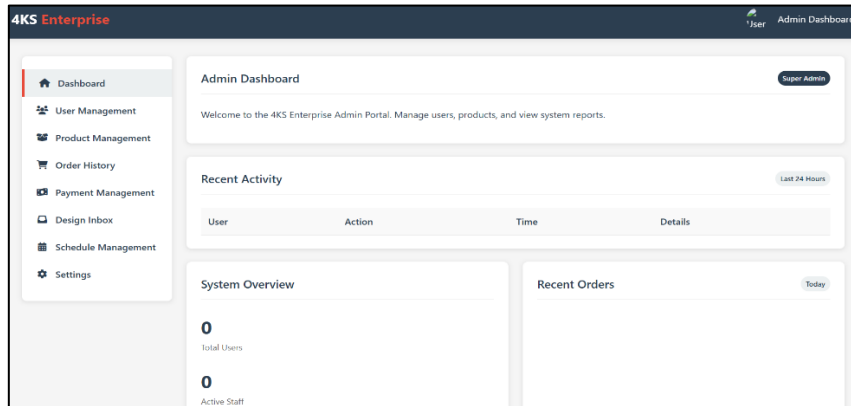


Figure 5. Dashboard for Web-based Ordering and Reservation System

Figure 6 shows the User Management interface of the 4ks Enterprise system, featuring a searchable table of user accounts with filters for roles and statuses. Administrators can review user details, track activity, and perform actions, with pagination for navigating through the list.

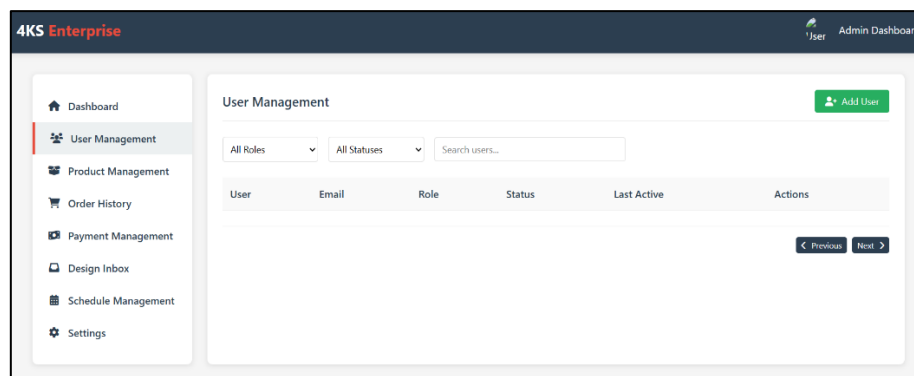


Figure 6. User Management for Web-based Ordering and Reservation System

Figure 7 shows the Product Management interface of the 4ks Enterprise system, featuring a searchable table of products with filters for categories and statuses. Administrators can monitor inventory, pricing, and availability, with options to modify entries.

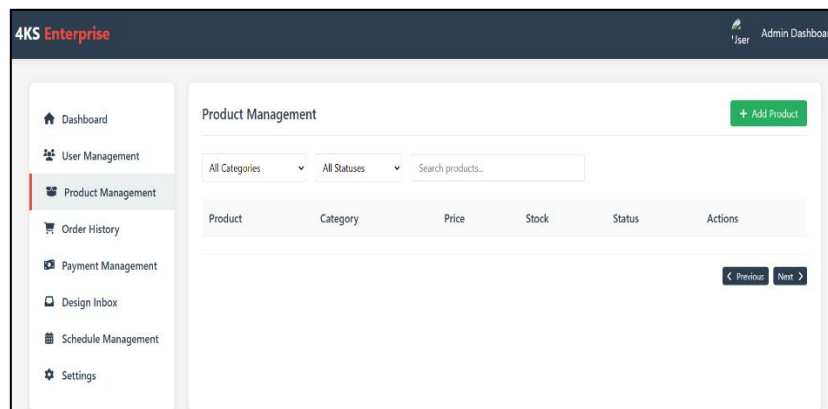


Figure 7. Product Management for Web-based Ordering and Reservation System

Figure 8 shows the Order History page of the 4ks Ordering and Reservation System allows users to manage Orders, view orders, delete and edit orders.

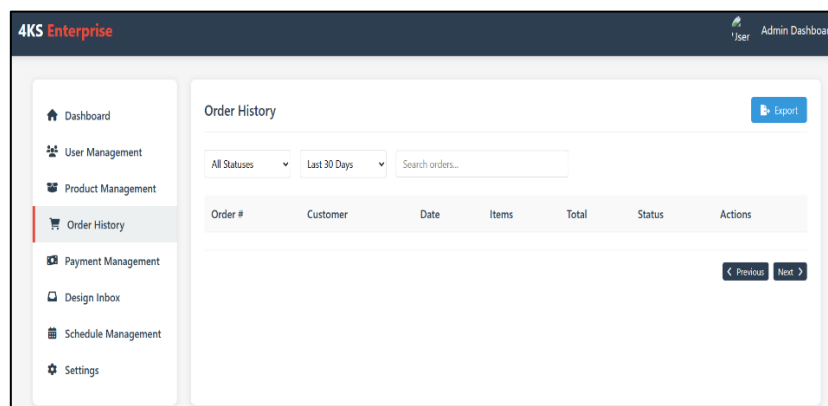


Figure 8. Order History for Web-based Ordering and Reservation System

Figure 9 shows the Payment Management section of the 4ks Enterprise Dashboard. This interface displays a table of payment records, including columns for Reference Number, Customer, Phone, Amount, Date, Receipt, Status, and Actions.

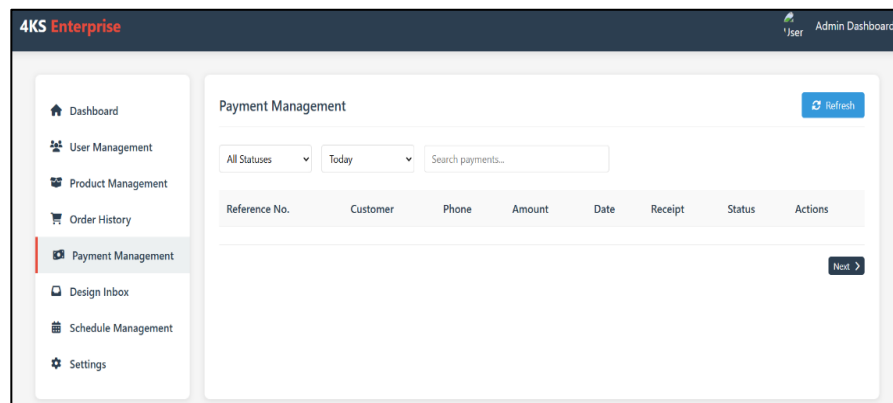


Figure 9. Payment Record for Web-based Ordering and Reservation System

Figure 10 shows the Design Inbox section of the 4ks Enterprise Dashboard, displaying a table of design orders with columns for Order number, Customer, Files, Submitted date, Status, and Actions.

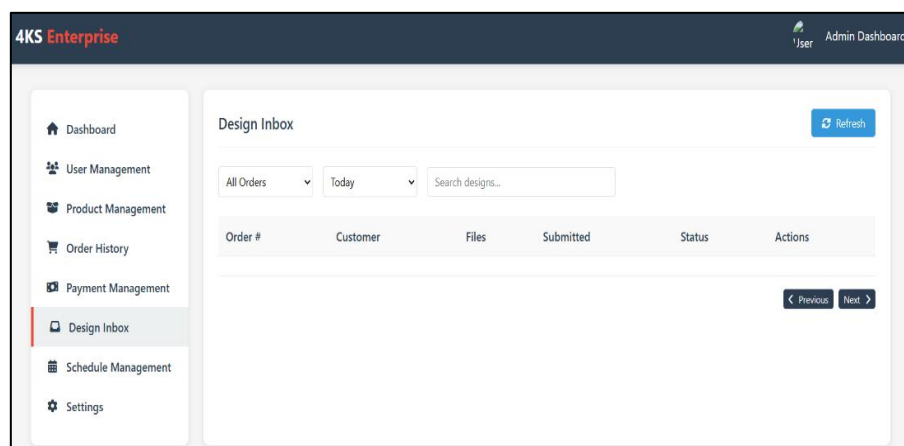


Figure 10. Admin Design Inbox Page for Web-based Ordering and Reservation System

Figure 11 shows the Schedule Management section, featuring a table of scheduled tasks with columns for Order number, Customer, Type, Date & Time, Location or Address, Status, and Actions.

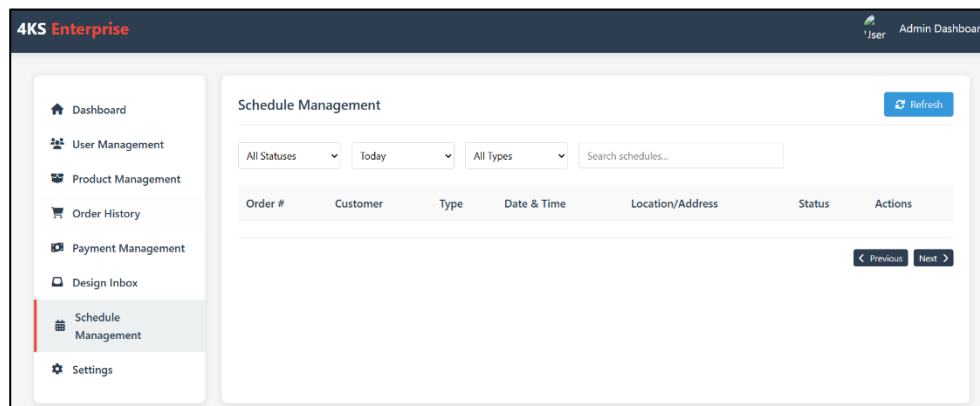


Figure 11. Admin Schedule Management Page for Web-based Ordering and Reservation System

Staff User Interface Design

Figure 12 shows a Staff Home page with sections for Recent Orders such as listing Order number, Customer, Items, Status, and Statistics.

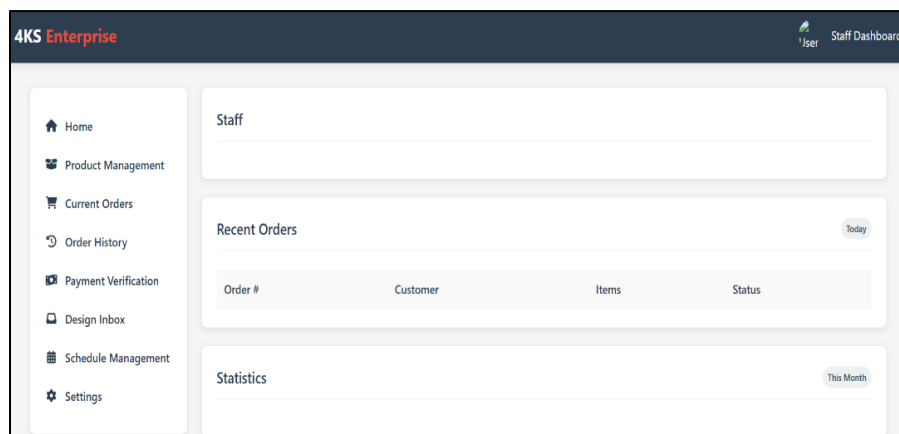


Figure 12. Staff Homepage for Web-based Ordering and Reservation System

Figure 13 shows the Product Management interface, allowing users to browse products by Category, Price, Stock, Status, and Actions. Options include "Add Product". A search bar and "All Categories" filter are provided.

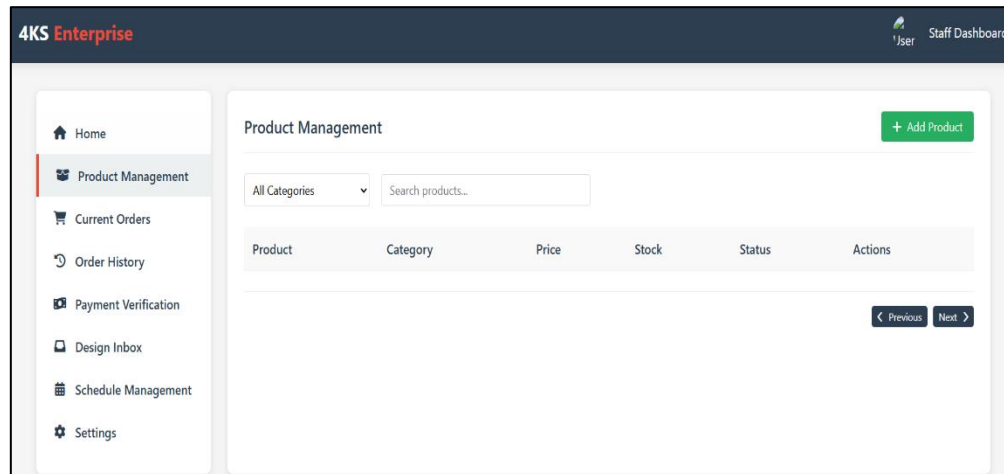


Figure 13. Staff Project Management for Web-based Ordering and Reservation System

Figure 14 shows the Current Orders section, listing active orders with details like Order number, Customer, Date, Items, Status, Total, and Actions. Filters include "All Statuses" and a search function.

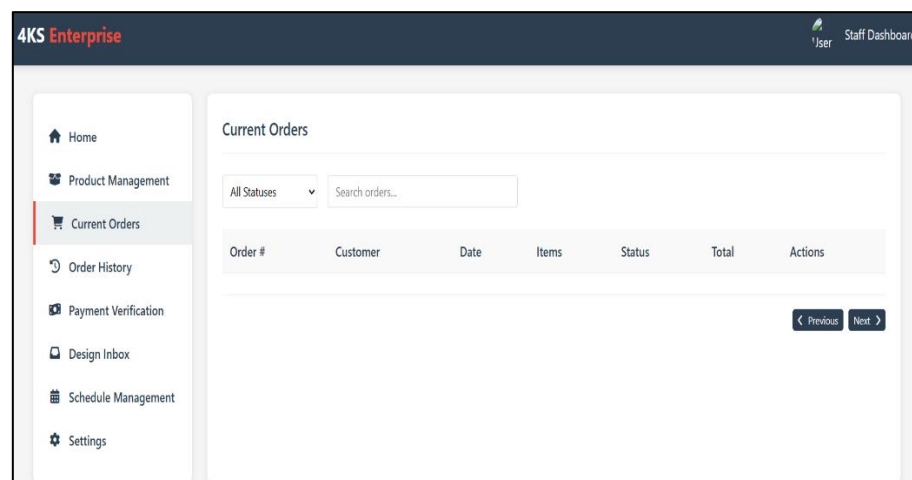


Figure 14. Staff Current Order page for Web-based Ordering and Reservation System

Figure 15 shows the Order History page, which displays past orders with columns for Order number, Customer, Date, Items, Status, Total, and Actions. Filters include "Last 30 Days" and "All Status."

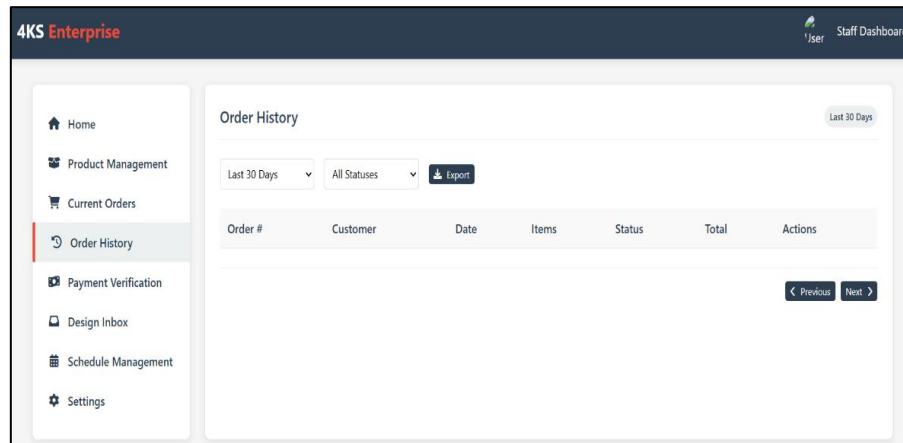


Figure 15. Staff Order History page for Web-based Ordering and Reservation System

Figure 16 shows the Payment Verification module, featuring a table of payments with Order number, Customer, Contact, Reference number, Amount, Payment Date, Receipt, Status, and Actions. Filters include "All Status" and search.

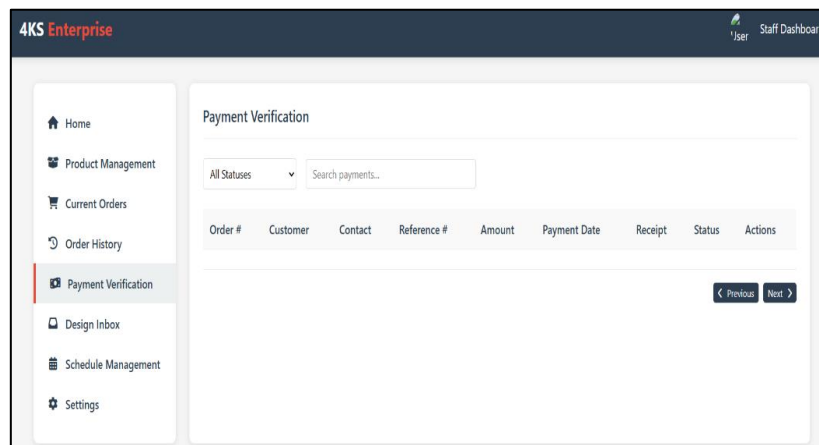


Figure 16. Staff Payment record page for Web-based Ordering and Reservation System

Figure 17 shows the Design Inbox with a table of design requests with the Order number, Customer, Product, Files, Status, Date Received, and Actions.

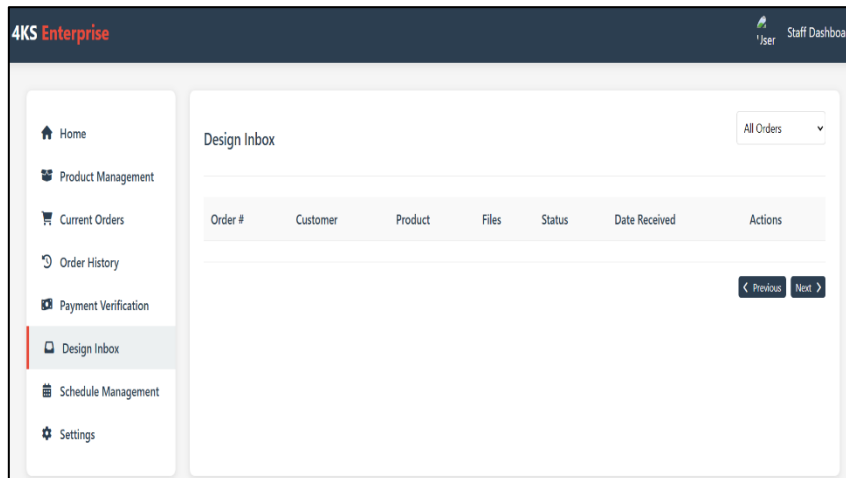


Figure 17. Staff Design Inbox page for Web-based Ordering and Reservation System

Figure 18 shows the Schedule Management table, listing appointments or tasks with the Order number, Customer, Type, Date & Time, Contact, Status, and Actions.

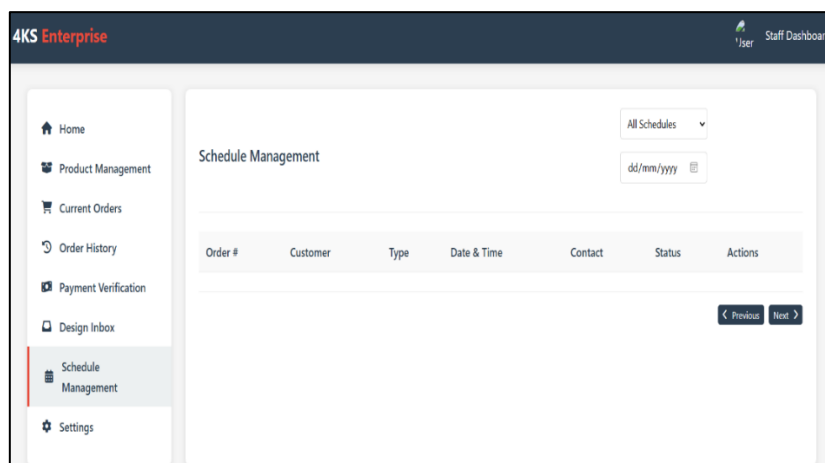


Figure 18. Staff Schedule Management for Web-based Ordering and Reservation System

Customer User Interface Design

Figure 19 shows the home page of the 4ks Enterprise system, displaying a welcome message for user "Kyle" along with sections for Active Orders, Recent Orders, and Popular Products.

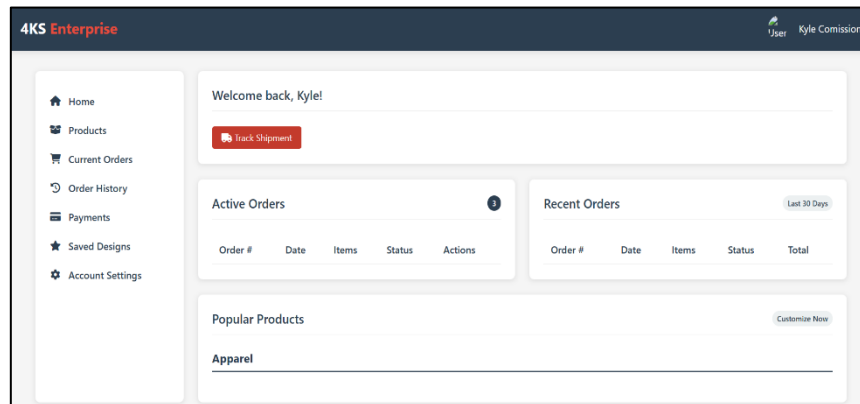


Figure 19. Customer Home page for Web-based Ordering and Reservation System

Figure 20 shows the products page of the 4ks Enterprise system, allowing users to search for products and browse the Apparel category, which is sorted by popularity by default.

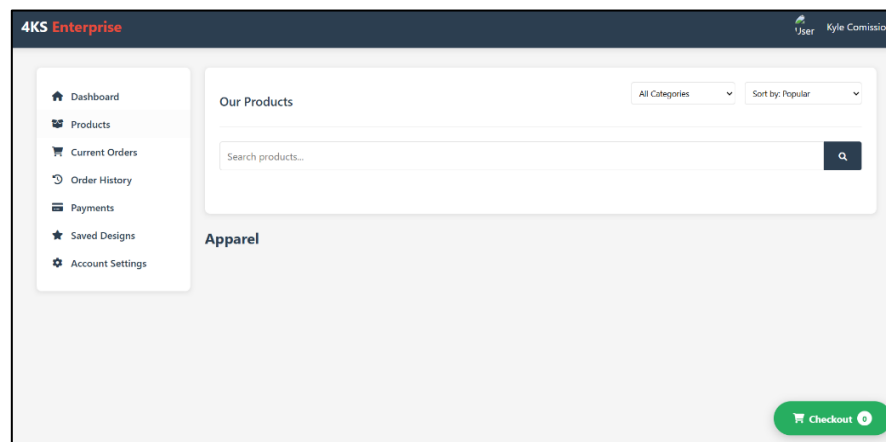


Figure 20. Customer Browse Product page for Web-based Ordering and Reservation System

Figure 21 shows the current orders page of the 4ks Enterprise system, featuring a filterable table of active orders with details like order number, date, items, status, and estimated delivery date, along with an order timeline section.

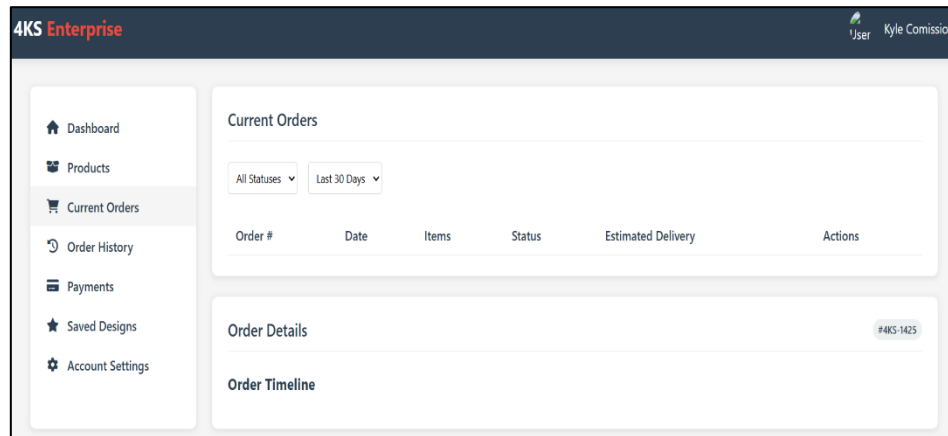


Figure 21. Customer Current Order page for Web-based Ordering and Reservation System

Figure 22 shows the order history page of the 4ks Enterprise system, displaying past orders with filters for time periods and a table showing order numbers, dates, items, statuses, totals, and actions.

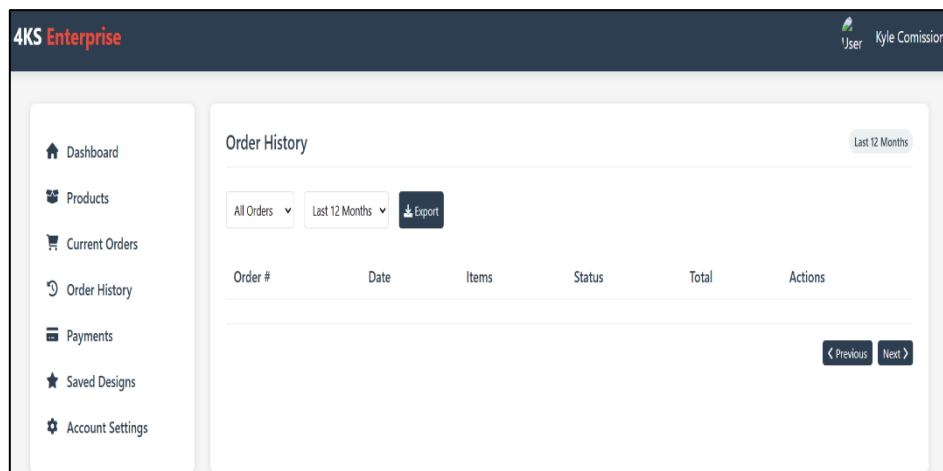


Figure 22. Customer Order History page for Web-based Ordering and Reservation System

Figure 23 shows the payment information page of the 4ks Enterprise system, detailing payment instructions, pending payments with due dates and amounts, payment methods like Gcash, and a history of past transactions.

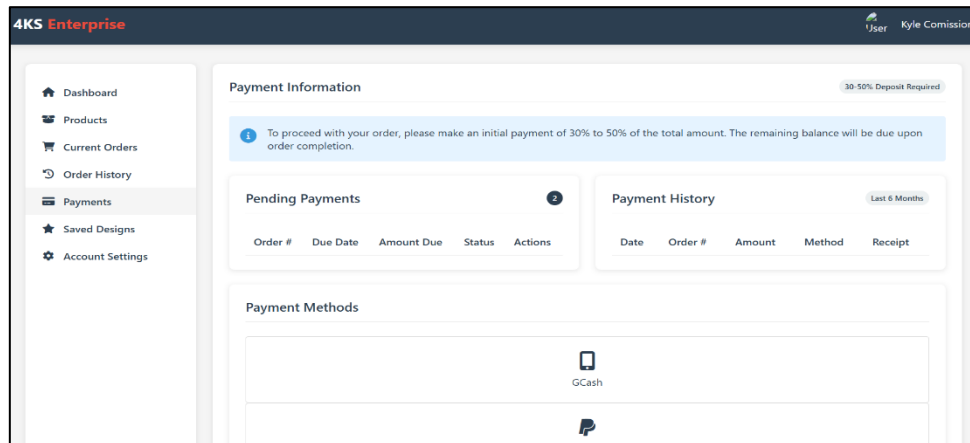


Figure 23. Customer Payment method page for Web-based Ordering and Reservation System

Figure 24 shows the saved designs page of the 4ks Enterprise system, providing options to filter designs by category or recency, search for specific designs, and create new designs.

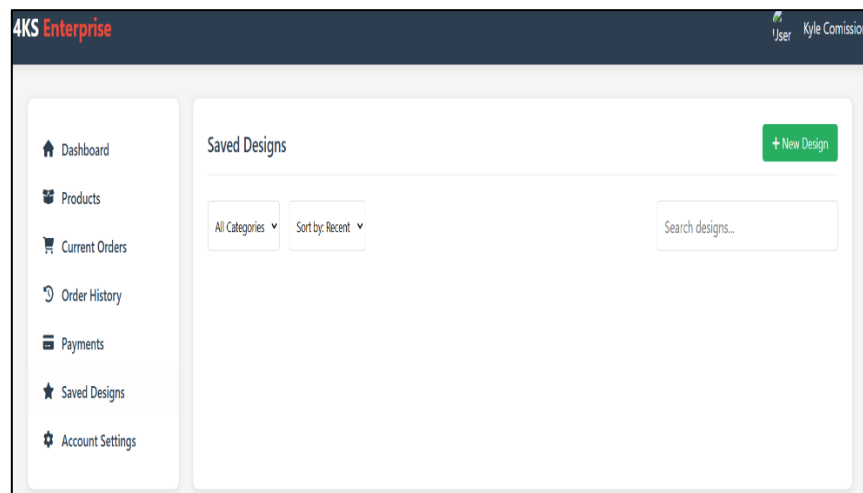


Figure 24. Customer Saved Design page for Web-based Ordering and Reservation System

Entity Relationship Diagram

Figure 25 illustrates how the different tables in the database are connected using one-to-one and one-to-many relationships to support the core operations of the system. At its core, the Customer manages Orders, which involve Order Details linked to Products and optional Designs, and generates a Payment. The Customer also maintains a Cart containing Cart_items and can make a Reservation, which is assigned to Staff. Both the Customer and Staff receive Notification messages, completing the structure for managing customers, orders, custom designs, payments, scheduling, and communication.

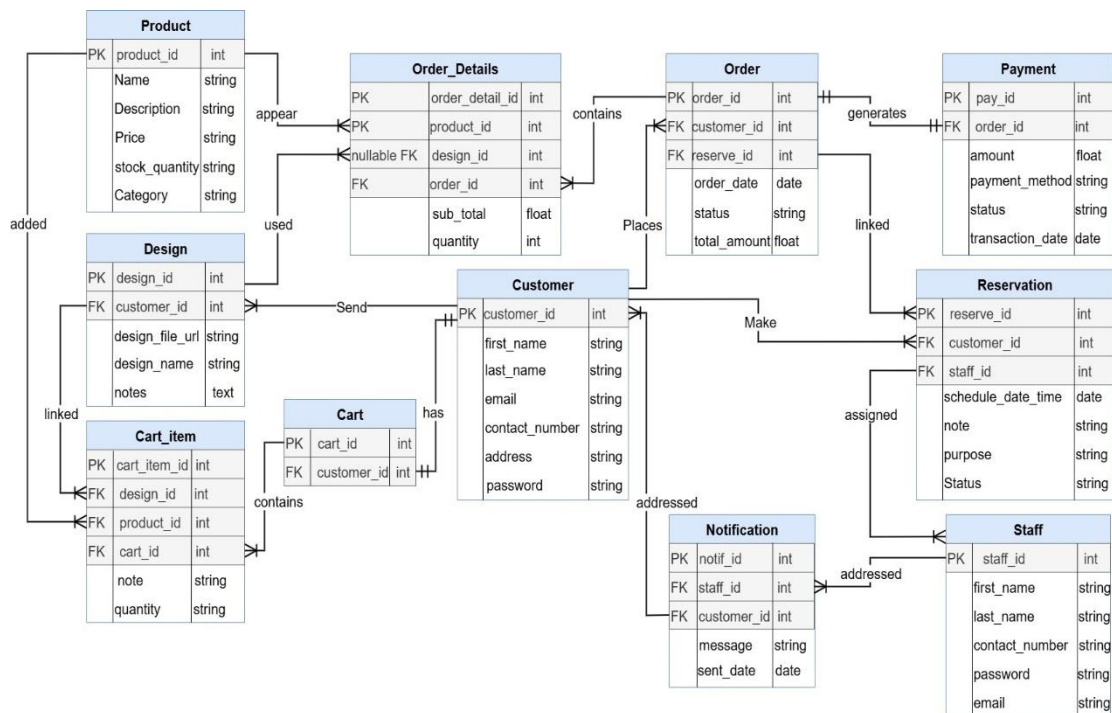


Figure 25. Entity Relationship Diagram

Use Case Diagram

Figure 26 shows the Use Case Diagram. the functional requirements of a system, involving three actors: Customer, Staff, and Owner. The Customer is responsible for system entry and core ordering processes, including the ability to register an account, Log In, and log out. Their primary functions revolve around Make an Order and Make a Reservation. an Order is extended by the ability to Upload Design Files and includes steps to browse products and place an Order. The Make Reservation use case includes Select Date. The customer also interacts directly with Make Payment and can access the Dashboard. The Staff actor shares access to Log In and Log Out. Their primary responsibilities include back-office operations: Manage Orders, Manage Schedule, Manage Payments, and Manage Product. The Owner actor has the highest level of control, sharing Log In and Log out, but is also responsible for Manage Inventory, Manage User Account, and Generate Reports. The Owner can also extend the functionality of Manage Product to cover all viewing and editing tasks.

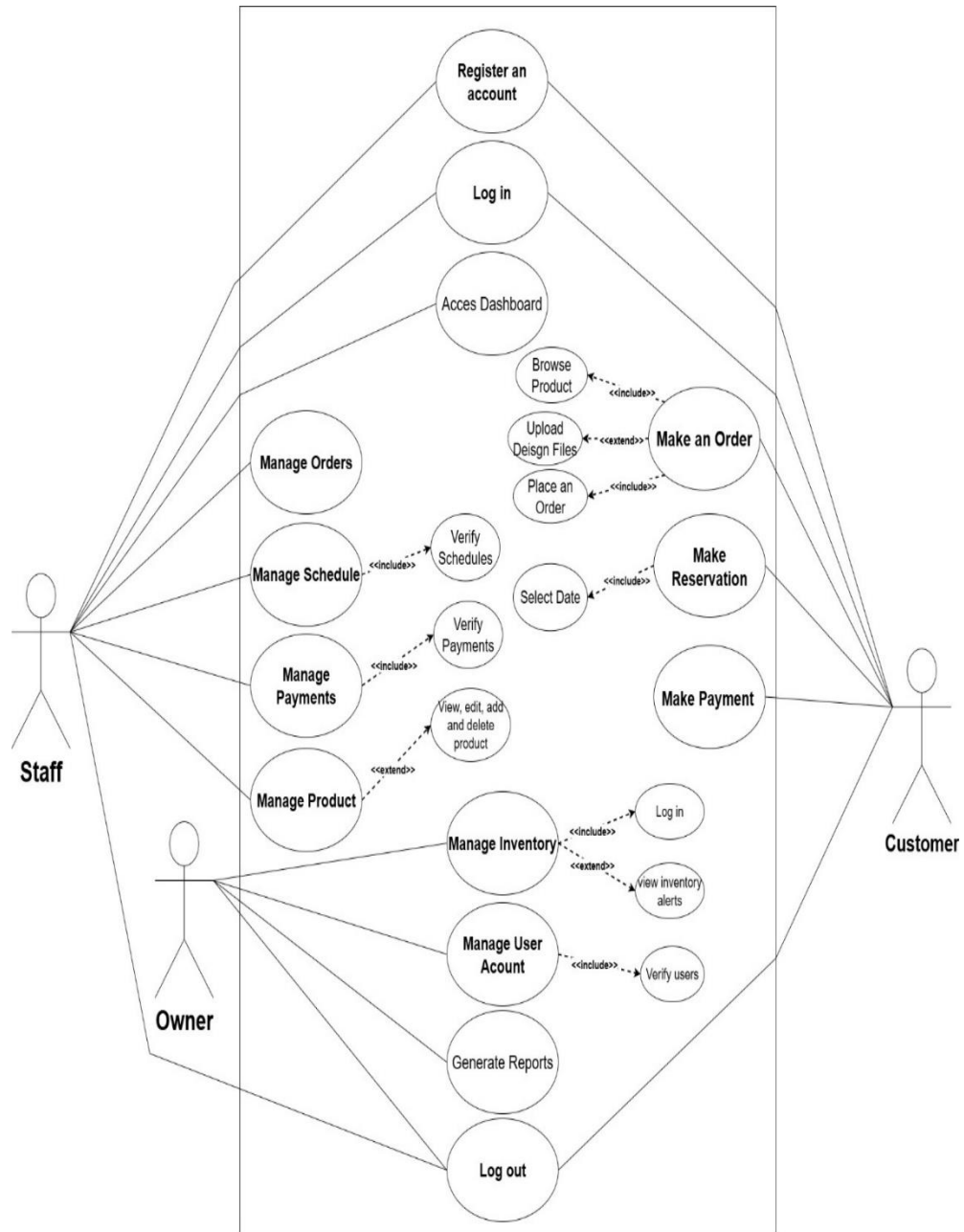


Figure 26. Use Case Diagram

Class Diagram

Figure 27 shows how the class diagram outlines a system structure based on the central User class, which is specialized into Admin, Staff, and Customer. The Customer places an Order, which is detailed by OrderDetail linking to Product and can include a DesignFile. The Order is processed through a Payment. Staff members are assigned to manage orders, reservations, and inventory. Admin users handle high-level management like user accounts, reporting, and setting inventory thresholds. All User types receive Notification objects for communication.

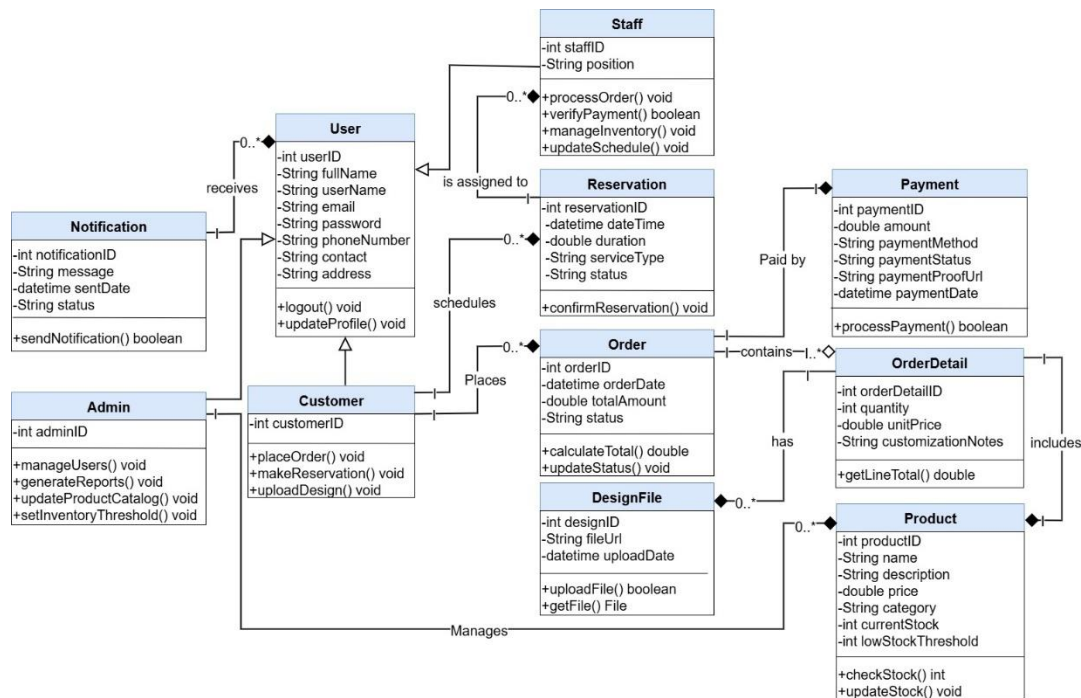


Figure 27. Class Diagram

Activity Diagram

Figure 28 shows the Customer Registration & Login Process, which begins with the System presenting the Registration form. The Customer is then required to fill out the registration form and click the register button. The System immediately checks if all required fields are filled and if the email is not already registered. If the data is not valid, the customer is redirected back to the Registration form. If the data is valid, the System proceeds to create a new customer record, display "Registration complete", and subsequently redirects to the login page. Once on the login page, the customer enters their username and password and clicks the "login" button. The System then checks if the credentials match a stored record. If the credentials do not match, the customer must re-enter them. If the credentials match, the System displays the customer dashboard, marking the successful end of the process.

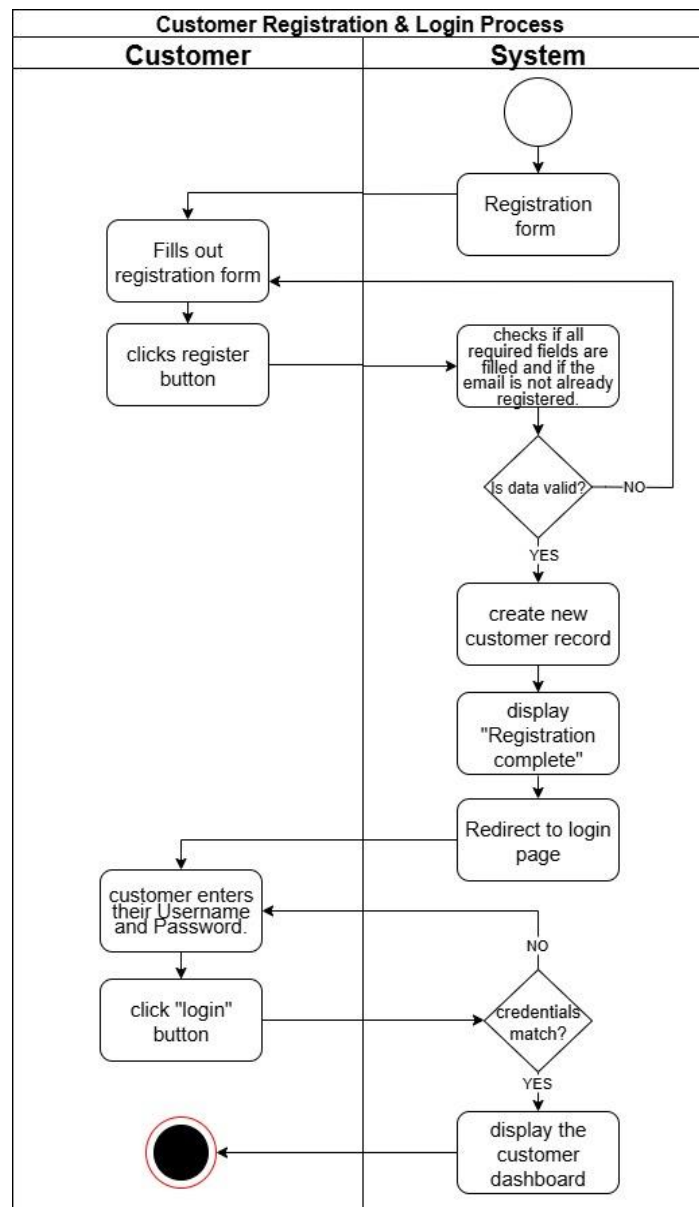


Figure 28. Activity Diagram for Customer Registration & Login Process

Figure 29 shows how the Customer Order Management Process begins with the Customer being logged in and the System displaying the product catalog. The customer selects items, specifies details, and adds them to the cart, which the System validates and updates. The customer then chooses to either continue shopping or proceed to checkout. At checkout, the System displays the shopping cart, and the customer enters the delivery address and preferred delivery method. Based on the fulfillment method chosen, such as pick up requires selecting a date or time slot, the process moves to payment. The System calculates the final cost, and the customer chooses payment if they pay in full or a 50% deposit. After the payment choice is registered with a pending status for 50% deposits, the System sends an order confirmation via email, and the process ends.

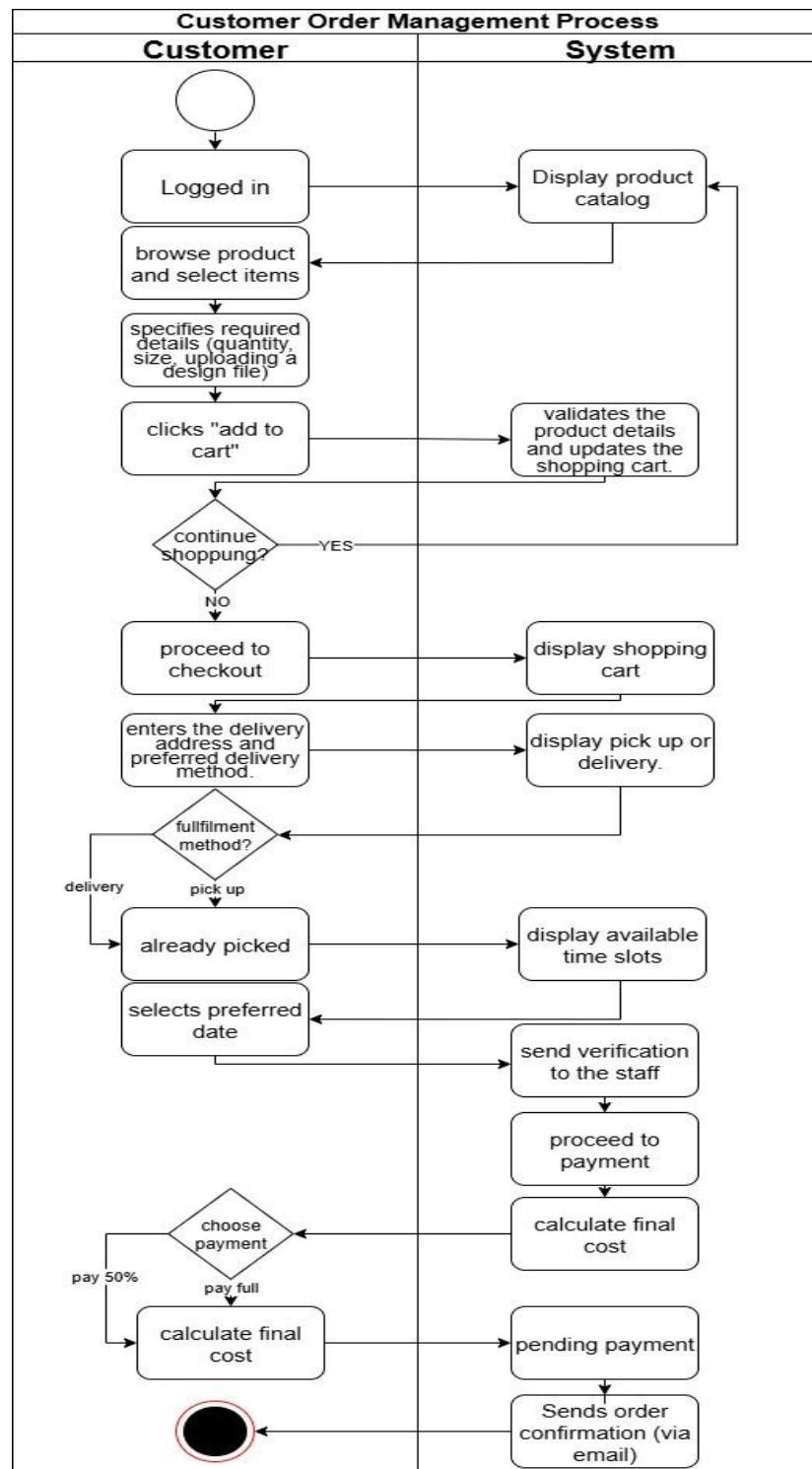


Figure 29. Activity Diagram for Customer Order Management

Figure 30 shows how the Customer Payment Process begins once the Customer is done placing an order. The customer proceeds to go to the "Payment" page, where they send a Screenshot Receipt of the Payment and then Input payment details. These details are received by the Staff, who then check the GCash account. If a Successful Payment is confirmed, the Staff will Confirm Payment, and the process concludes. If the payment is not successful, the Staff will send a message that the payment has not yet been sent. Upon receiving this message, the Customer is prompted to resend the payment, after which the process also ends.

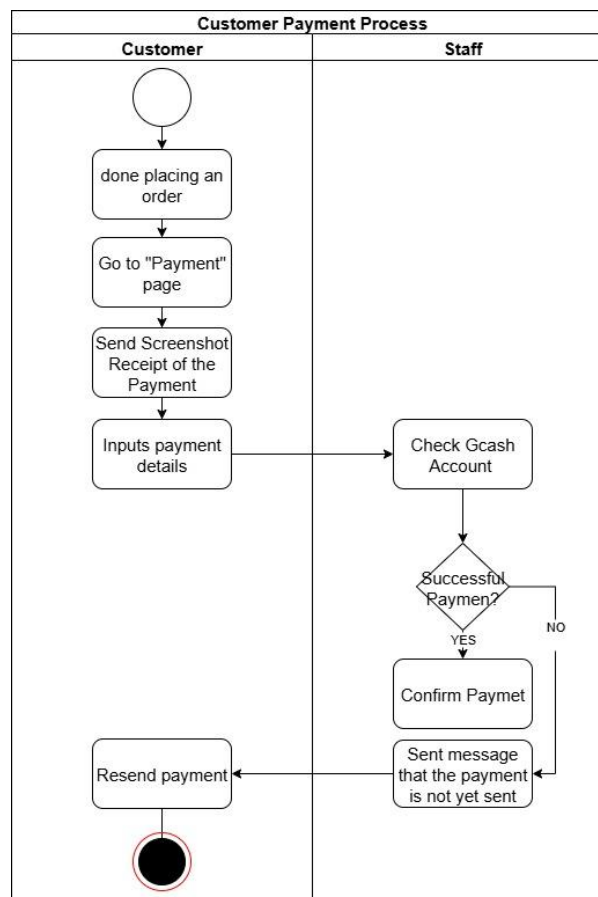


Figure 30. Activity Diagram of Customer Payment Process

Figure 31 shows that the Customer Upload Design File process starts with the Customer going to the saved designs page, which prompts the System to display the design upload interface. The customer then prepares the design, clicks the upload file button, and selects the file from their device. The System receives the file and checks for a valid format. If invalid, the process loops back to the upload interface. If valid, the System moves the validated file. The customer then confirms the design, and the System permanently links the uploaded file's unique ID to the current Order or Cart item, completing the process.

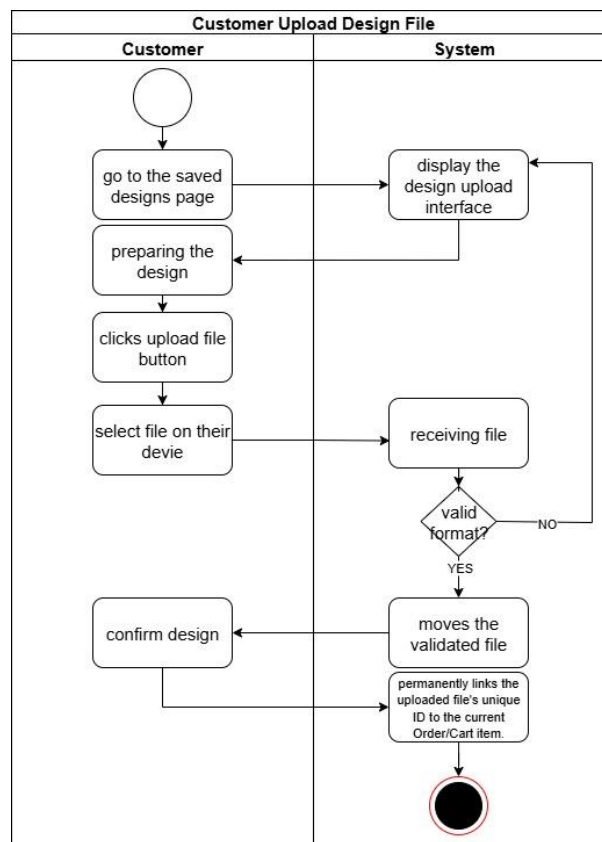


Figure 31. Activity Diagram Customer upload design file

Figure 32 shows that the Staff Registration process starts with the Admin logging into the dashboard, which prompts the System to display the Admin dashboard. The admin then navigates to user management. The System will display user management, then the Admin clicks “Add new staff,” and the system will display staff registration. After that, the Admin will fill in the staff details or data, and the system will validate the data. If it is not, it will go back to filling up the details, and if it is yes, it will create a staff account. The system will also send an email to staff with login information.

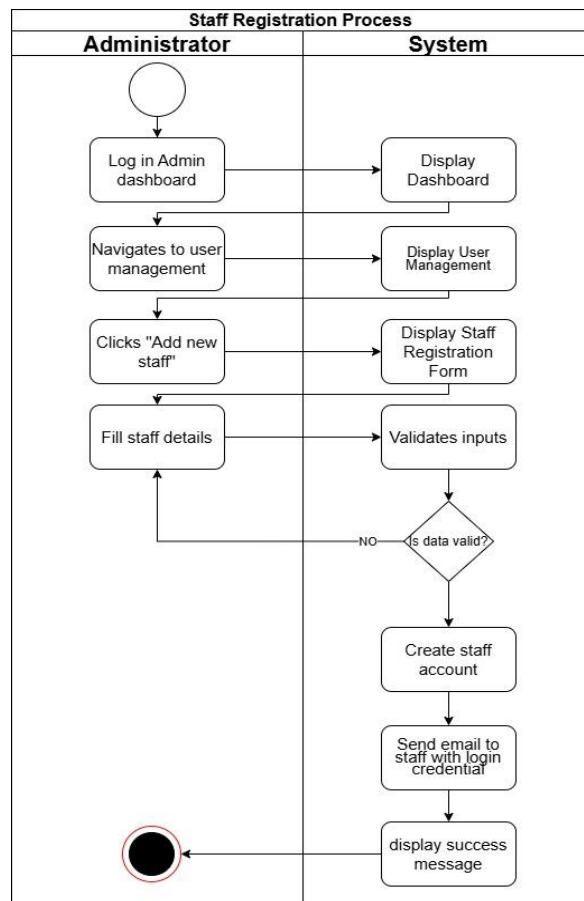


Figure 32. Activity Diagram Staff Registration

Figure 33 shows how the Staff Payment Verification process starts with the System indicating a Payment verification pending. The Staff accesses the verification page, selects an Order, and the System displays the Uploaded Proof of Payment. The staff then opens the GCash app to check for a matching transaction. If Payment is verified YES, the staff approves Payment, and the System updates the status to "paid" and automatically notifies the customer, ending the process. If NO, the staff rejects payment, and the System sends a clarification request, which loops back for a new verification attempt.

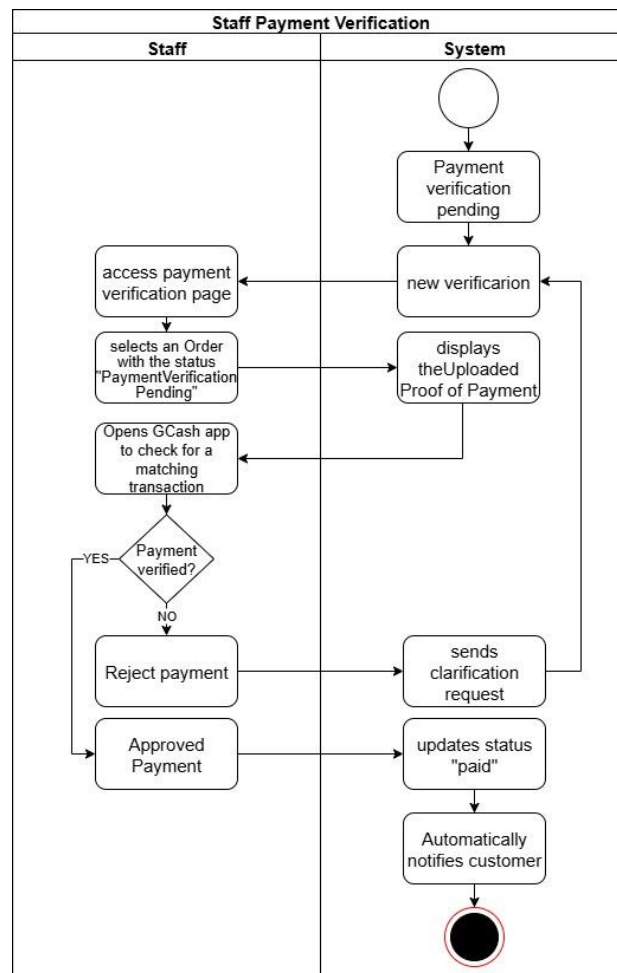


Figure 33. Activity Diagram of Staff Payment Verification Process

Figure 34 shows how the Staff Order Management Process begins with the Staff performing a Login action. Once logged in, the staff member proceeds to Open [the] Order dashboard, where they can view a list of orders. The staff member then selects an order from the list. The next critical step is to update the order status, choosing from options like Processing, complete, or cancelled. This action triggers the System to save an update. Finally, the System sends confirmation to the customer via email, bringing the entire order management process to an end.

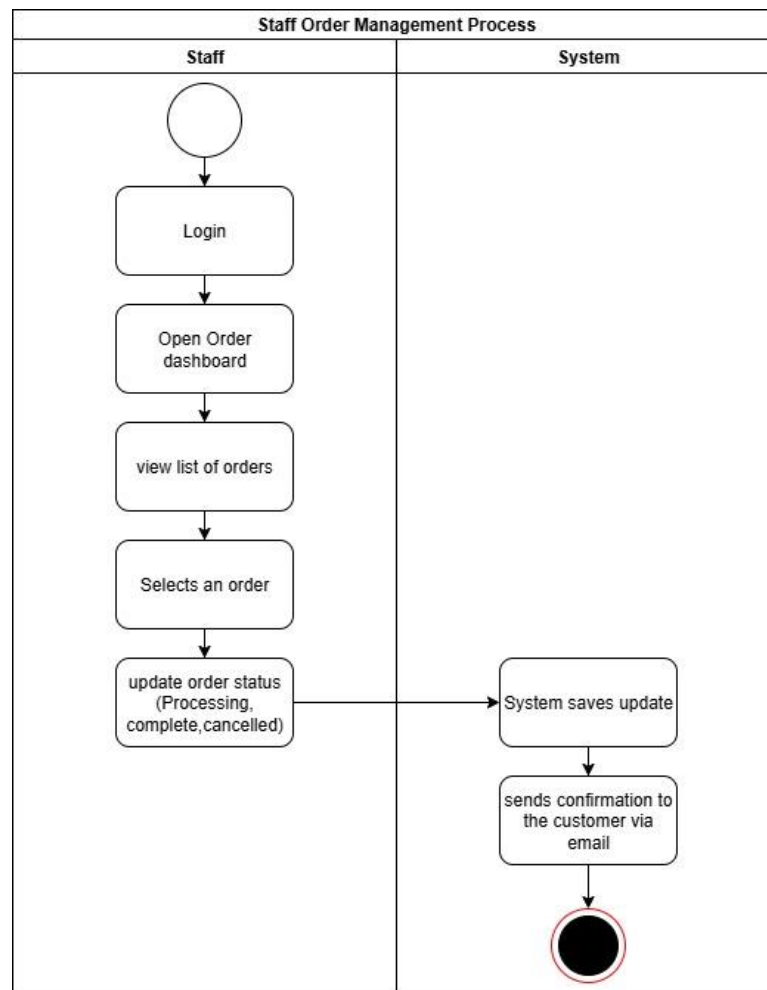


Figure 34. Activity Diagram of Staff Order Management Process

Figure 35. This shows the Staff Schedule Management process begins with the Staff performing a Login action. Once logged in, the staff member opens the schedule dashboard, which allows them to view schedule requests. The staff then selects a reservation/order from the list of requests. After selection, the staff member Updates or confirms the schedule. This action triggers the System to save update. Finally, the System Sends a notification regarding the schedule change, and the process concludes.

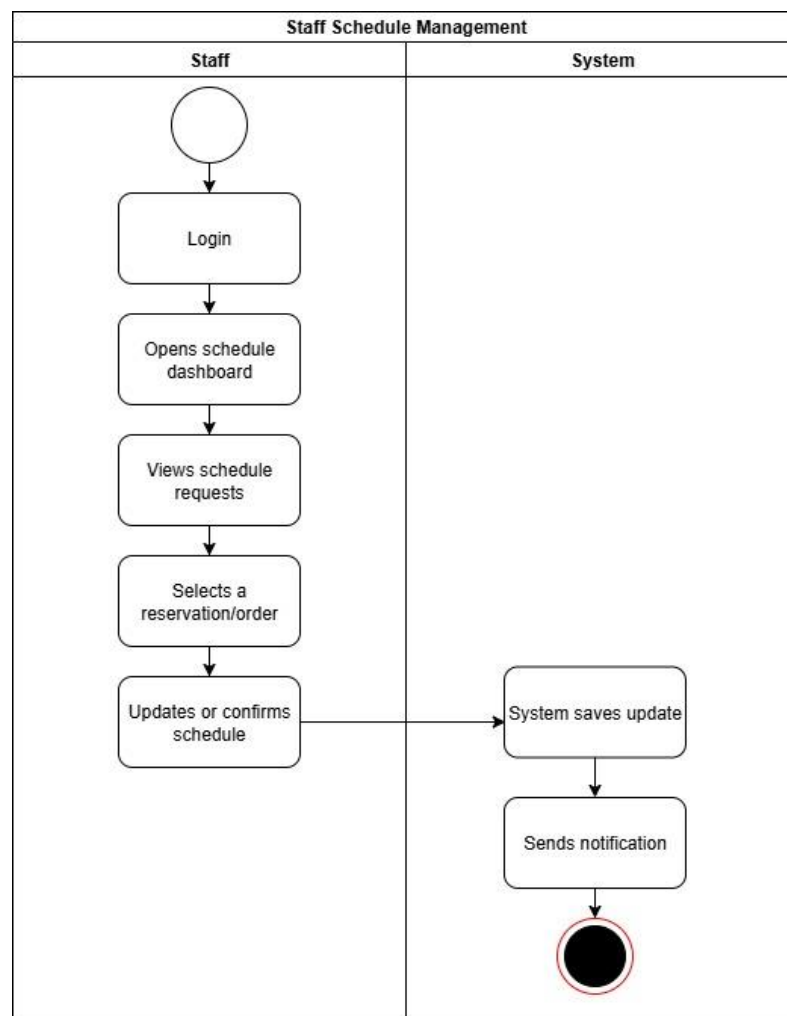


Figure 35. Activity Diagram of Staff Schedule Management

Figure 36 shows how the Staff Design Inbox Management process begins with the Staff performing a Login action. After logging in, the staff member Opens design inbox and proceeds to View uploaded design file. Next, the staff member Reviews design details. A decision point is then reached: Is design acceptable? If the design is acceptable YES, the staff member will Approve design. If the design is not acceptable NO, the staff member will reject design. Both outcomes lead to a final action by the System: if approved, the System updates status, if rejected, the System notifies customer. Both outcomes lead to a final action by the System: if approved, the System updates status, if rejected, the System notifies customer.

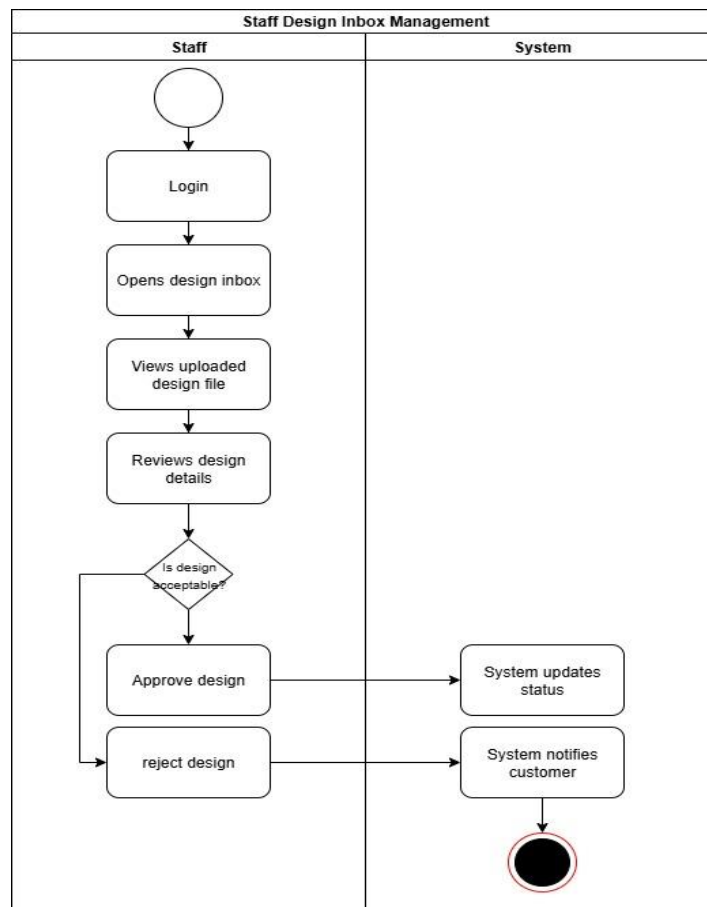


Figure 36. Activity Diagram of Staff Design Inbox Management

Figure 37 shows the Admin User Management process involves the Admin performing Registration and then Login to the System. If the System Validates information successfully, the Admin can Create an account, and the System confirms Success. If validation fails, an Error message is shown, requiring the Admin to re-login. After successful creation, the Admin accesses the 4ks Enterprise dashboard to manage users account. This action prompts the System Display Accounts, allowing the Admin to add, Edit and Delete User accounts before the System Displays profile and the process ends.

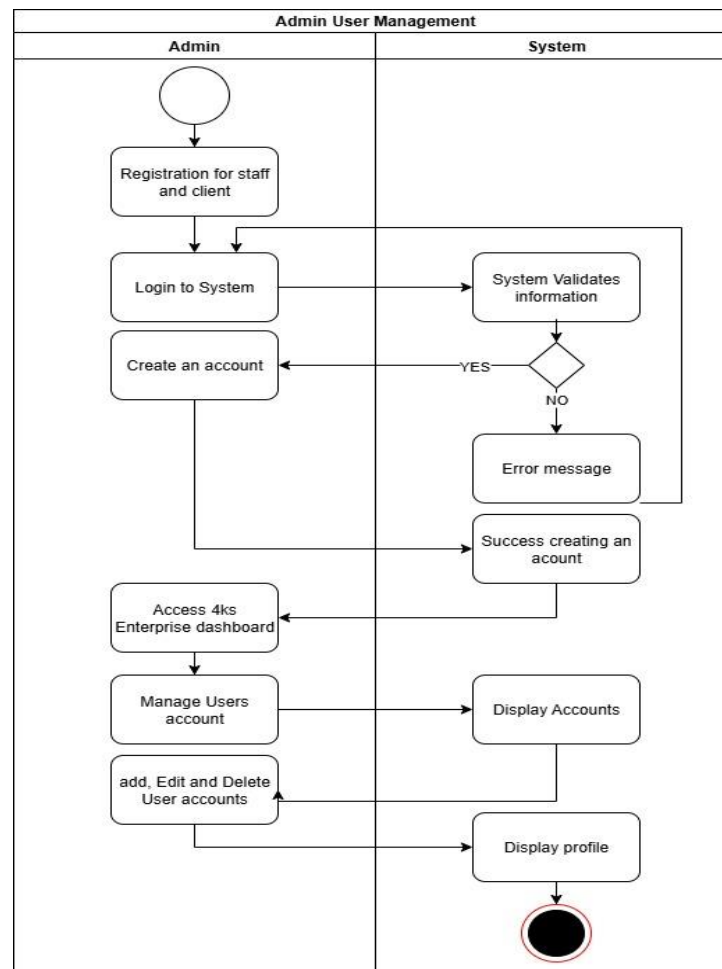


Figure 37. Activity Diagram of Admin User Management

Figure 38 shows how the Admin Inventory Management process begins with the Admin reviewing pending restock request. The Admin then proceeds to approve or reject orders based on budget before taking the action to place purchase orders with suppliers. Once the orders are placed, the Admin receives new stock. This triggers the System to update stock Record and then record the saved new Stocks. The Admin then adjusts inventory thresholds as needed, which prompts the System to apply new threshold. Finally, the System Check Threshold, and the inventory management process concludes.

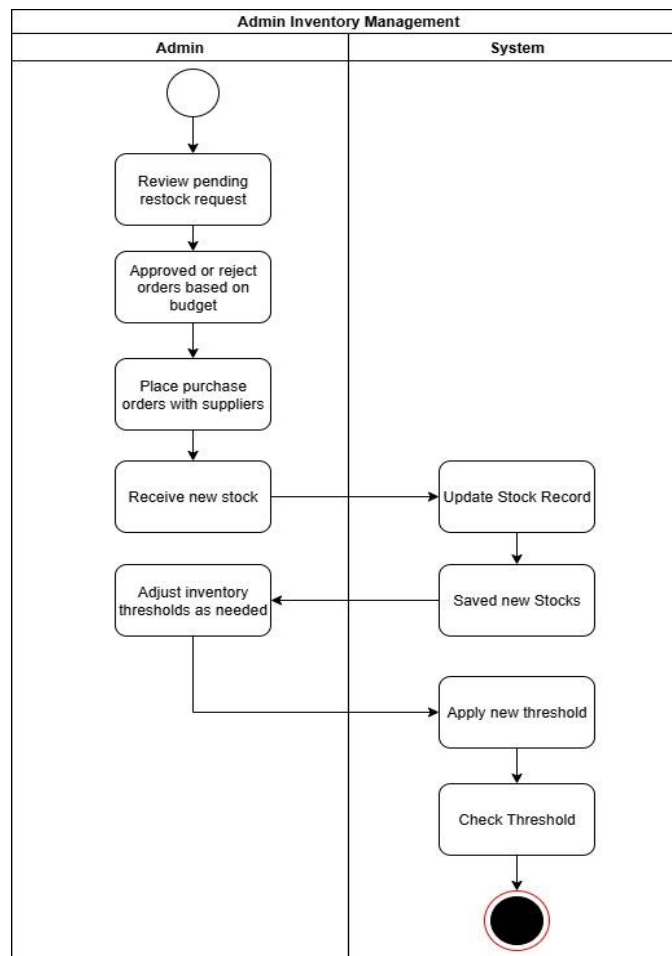


Figure 38. Activity Diagram of Admin Inventory Management

Figure 39 shows how the Admin Product Management process begins with the admin initiating the process to Open Product Management. Once opened, the user proceeds to view the product list. The user then performs the core function of the process: Add/edit/delete products. This action triggers the System to update the database. After the update is complete, the System provides a Confirmation message, at which point the entire product management process concludes.

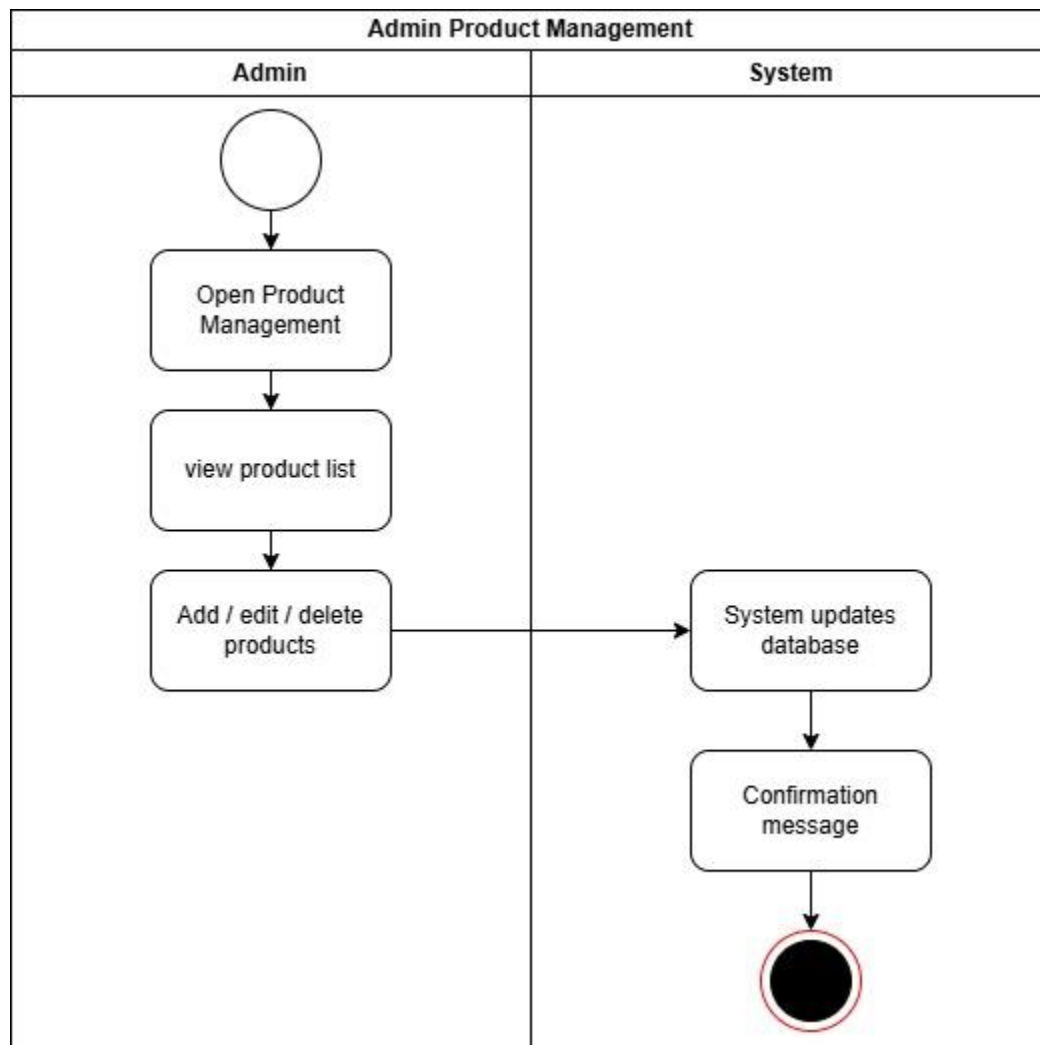


Figure 39. Activity Diagram of Admin Product Management

Figure 40 shows how the Staff and Admin Report Generation Process, which shows how users create different types of reports with the help of the system. The process begins when the staff or admin logs in and opens the reporting module. From there, they select the type of report they want, such as sales, payments, or inventory. Once the report type is chosen, the system gathers the necessary data and generates the report. After the report is created, the user can view it directly or download it for further use.

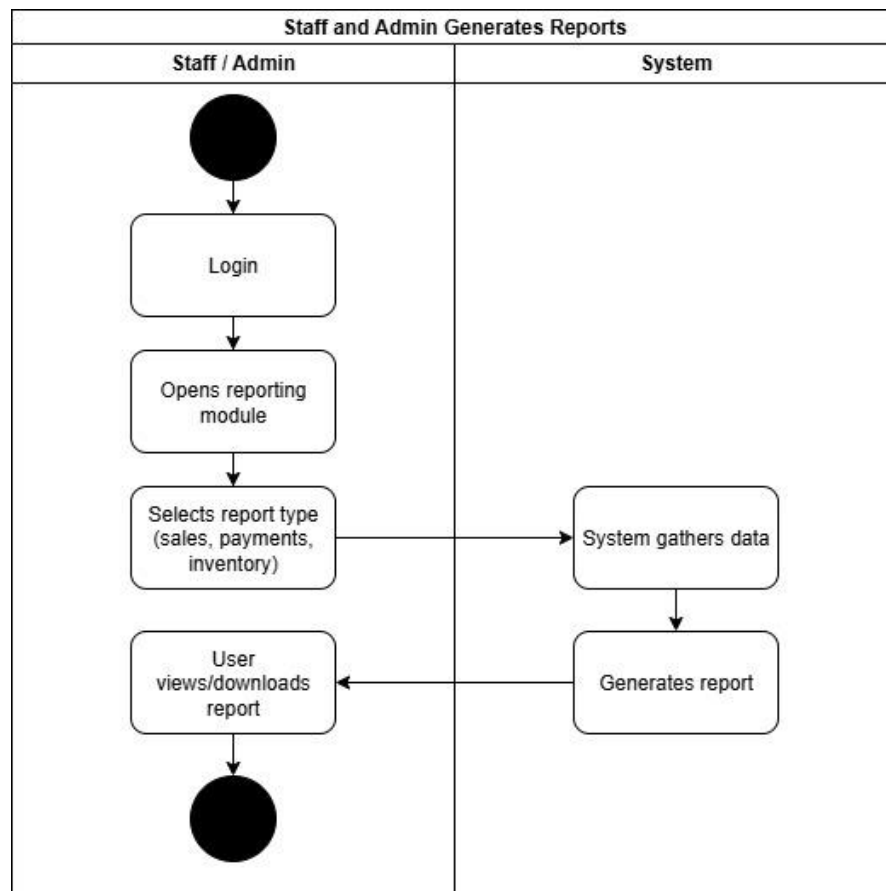


Figure 40. Activity Diagram of Staff and Admin Generate Reports

Project Cost Estimation

The table 4 illustrates the estimated project cost of the Web-Based Ordering and Reservation System.

Table 4. Project Cost Estimate

Items		Estimated Cost (PHP)
1	Requirements Analysis	
	a. Snack/Food	300.00
	b. Transportation	200.00
	c. Research Materials	500.00
	d. Printing Documents	1000.00
	e. Internet Connectivity	1,100.00
2	System Design	
	a. Interface Design	500.00
	b. Backend Design	300.00
	c. Design Database	300.00
	d. Specifications	300.00
3	Development	
	a. Design Specifications	300.00
	b. Integrate Modules	250.00
	c. Design Specifications	250.00
4	Implementation	
	a. System Testing	200.00
	b. Correct Issues	200.00
Total		5,700.00

Gantt Chart

Figure 41 presents the Gantt chart for the project, outlining a structured timeline from March 2025 – December 2025. The plan is organized into critical phases. Including Requirement Analysis, System Design, Development, Testing, Implementation.

ID	Task	Start Date	Due Date	Duration
1	Requirement Analysis	03/03/2025	19/03/2025	17 days
1.1	Define Problem	03/03/2025	07/03/2025	5 days
1.2	Define Scope	08/03/2025	11/03/2025	4 days
1.3	Define Expected Outcomes	12/03/2025	14/03/2025	3 days
1.4	Define Objectives	15/03/2025	19/03/2025	5 days
2	System Design	20/03/2025	18/04/2025	29 days
2.1	Modeling Diagrams	20/03/2025	29/03/2025	10 days
2.2	Hardware and Software Specifications	30/03/2025	05/04/2025	6 days
2.3	User Interface Design	06/04/2025	18/04/2025	13 days
3	Development	05/08/2025	20/08/2025	16 days
3.1	Production Development	05/08/2025	13/08/2025	9 days
3.2	User Training	14/08/2025	20/08/2025	7 days
4	Testing	01/10/2025	12/11/2025	38 days
4.1	Unit Testing	25/08/2025	01/09/2025	8 days
4.2	Integration Testing	02/09/2025	11/09/2025	10 days
4.3	System Design	12/09/2025	23/09/2025	12 days
4.4	User Acceptance Testing	24/09/2025	01/10/2025	8 days
5	Implementation	05/11/2025	03/12/2025	2 months
5.1	Frontend and Backend Design	05/11/2025	18/11/2025	45 days
5.2	Database Design	19/11/2025	03/12/2025	15 days

Figure 41. Gantt Chart for Web-Based Ordering and Reservation System